DEEP KERNELIZATION FOR THE TREE BISECTION AND RECONNECTION (TBR) DISTANCE IN PHYLOGENETICS

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ABSTRACT. We describe a kernel of size $9k-8$ for the NP-hard problem of computing the Tree Bisection and Reconnection (TBR) distance k between two unrooted binary phylogenetic trees. To achieve this, we extend the existing portfolio of reduction rules with three new reduction rules. Two of these are based on the idea of topologically transforming the trees in a distance-preserving way in order to guarantee execution of earlier reduction rules. The third rule extends the local neighborhood approach introduced in [20] to more global structures, allowing new situations to be identified when deletion of a leaf definitely reduces the TBR distance by one. The bound on the kernel size is tight up to an additive term. Our results also apply to the equivalent problem of computing a maximum agreement forest between two unrooted binary phylogenetic trees. We anticipate that our results are widely applicable for computing agreement-forest based dissimilarity measures.

1. INTRODUCTION

A phylogenetic tree is essentially a tree in the usual graph-theoretical sense whose leaves are bijectively labeled by a set of labels X [26]. Such trees have a central role in the study of evolution. The label set X represents a set of contemporary species, the unlabeled interior vertices of the tree represent hypothetical (extinct) ancestors of X and the topology of the tree encodes the history of branching events, such as speciation, which caused those ancestors to diversify into the set of species X. A central challenge in the field of phylogenetics is to accurately infer such trees from data obtained solely from X, such as DNA data [10]. However, it is not uncommon to obtain different trees for the same set X; this can be methodological (e.g. different objective functions or multiple optima) or due to the fact that some species have multiple distinct tree signals woven into their genome [27]. This motivates the use of distance measures in phylogenetics, which rigorously quantify the dissimilarity of two phylogenetic trees. Such distance measures can communicate important information about the biological significance of the observed differences [32] and can help us to understand the behavior of tree-construction algorithms that traverse the space of phylogenetic trees by applying local rearrangement operations [18, 23]. Distances can also be used as part of the toolkit for constructing non-treelike hypotheses of evolution, known as phylogenetic networks [17].

In this article we are concerned with one such distance, the Tree Bisection and Reconnection (TBR) distance, which is a metric on the space of unrooted (i.e. undirected) binary

Date: January 30, 2024.

Key words and phrases. phylogenetics, agreement forest, TBR distance, kernelization, fixed parameter tractability.

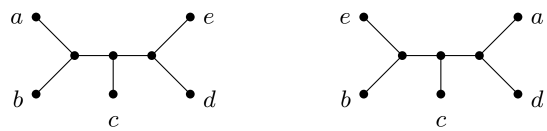

FIGURE 1. Two unrooted binary phylogenetic trees on $X = \{a, b, c, d, e\}$. Note that the TBR distance (formally defined in Section 2.1) between these two trees is two.

phylogenetic trees. The TBR distance $d_{\text{TBR}}(T, T')$ between two unrooted binary phylogenetic trees T and T' on the same leaf set represents the minimum number of TBR operations required to transform T into T'. Each operation detaches a subtree of one tree and reattaches it elsewhere. An example of a TBR operation is shown in Figure 2. See Figure 1 for two unrooted binary phylogenetic trees whose TBR distance is two. It is NP-hard to compute $d_{\text{TBR}}(T, T')$ [2, 15]. The problem has an equivalent, alternative formulation using *agreement forests*. An agreement forest is a partition of X such that the spanning trees induced by the blocks of the partition are disjoint in both trees *and* the induced spanning trees have the same topology in both trees, up to suppression of degree-2 vertices. An agreement forest with a minimum number of blocks is called a *maximum agreement forest* (MAF); it is well-known that the TBR distance (d_{TBR}) is equal to the number of blocks in a MAF (d_{MAF}) minus 1 [2]. In the last twenty years, maximum agreement forests have received sustained attention from the mathematics, computer science (e.g., in parameterized algorithmics) and bioinformatics communities, see e.g. [2, 3, 7, 9, 22, 24, 25, 31]. One response to the NP-hardness of computing d_{TBR} is *kernelization*. Here the goal is to apply polynomial-time preprocessing rules such that d_{TBR} is preserved, or decreased in a controlled fashion, such that the reduced trees have at most $f(d_{\text{TBR}})$ leaves for some function f that depends only on d_{TBR}. For further background on kernelization we refer to the book [14]. The core idea is that, if d_{TBR} is small, then the reduced trees (known as the *kernel*) will be small even if $|X|$ is very large and d_{TBR} can be computed on these small trees using optimized exponential-time algorithms. The use of kernelization in this context is not coincidental: phylogenetics continues to be a rich source of open problems in, and application opportunities for, parameterized complexity [6].

In 2001 it was shown in [2] that the *subtree* and *chain* reduction rules suffice to obtain a kernel of size at most $28k$, where $k = d_{\text{TBR}}$. These function by reducing common pendant subtrees and common chains (i.e. caterpillar-like regions), respectively. Almost 20 years later the present authors proved that the same reduction rules actually yield a kernel of size at most $15k-9$, and in fact that this is tight [19]. A critical insight in [19] was that computation of d_{TBR} (or d_{MAF}) can equivalently be viewed as the problem of adding the labels X, and a set of *breakpoints* (essentially: edge cuts), to an (unknown) cubic multigraph, known as a *generator*, such that the original two trees can be retrieved (see Figure 3). This insight was subsequently leveraged in [20] to design five new reduction rules which, when added to the subtree and chain reduction rules, yield a tight kernel of size $11k-9$. An empirical follow-up showed that the new rules in [20] have added reductive power in practice [30], and recently the reduction rules have successfully been mapped and adapted to distances and agreement forests on *rooted* trees [21], enabling smaller kernels to also be obtained there.

Following the reduction in the size of the d_{TBR} kernel from $28k$ to $15k-9$, and then to $11k-9$, it is natural to ask: can we do better than $11k-9$? In this article we answer the question affirmatively: we give a kernel of size $9k-8$, which is tight up to an additive term. We note that such an ongoing research effort is certainly not unprecedented in the parameterized complexity literature. For example, in a sequence of articles the kernel for the (unrelated) Feedback Vertex Set problem on planar graphs was progressively reduced from $112k$ to $13k$, where k is the size of a feedback vertex set of the input [1, 4, 5, 33]. Our result fits in this tradition. Large portfolios of reduction rules can also be very powerful in practice. For example, the winning exact solver for the well-known Vertex Cover problem in the PACE 2019 algorithm engineering contest uses a wide array of reduction rules [16].

To obtain a kernel of size $9k-8$ we use the analytical and counting bottlenecks identified in [20] as a starting point, and use these to guide the design of three new reduction rules. The reduction rules have a very different flavor to what has come before. The first new reduction rule addresses the following bottleneck: some of the topological structures that contribute heavily to the $11k-9$ bound, and which we thus wish to target for reduction, could potentially be leveraged by a depth-bounded branching algorithm that recursively cuts edges in the input trees to obtain an agreement forest. However, the cuts applied by such a direct, non-preprocessing algorithm yield a different, more general problem, on forests rather than trees, which is analytically far harder to deal with from a kernelization perspective. The first new reduction rule, Reduction 8, circumvents this by applying a d_{TBR}-preserving transformation to one of the trees, such that the classical subtree reduction rule again becomes possible and the number of leaves can definitely be reduced overall. In other words, we jump "sideways" to an alternative pair of trees, with the same TBR distance, which is more amenable to earlier reduction rules. Notably, to make this sideways step, the reduction in some sense reverses the classical chain reduction, thereby lengthening another common chain very slightly. This reversal is not a goal in itself, but rather a convenient way of ensuring that the TBR distance between the two trees does not change. (We refer to [11] for related discussions on running reduction rules "backwards".) The second new reduction rule, Reduction 9, works by identifying other topological structures which contribute heavily to the $11k-9$ bound, and transforming them into structures that can be attacked by Reduction 8. Finally, Reduction 10 is similar in spirit to Reduction 9, but is more direct: if it triggers, it is parameter reducing i.e. d_{TBR} is definitely reduced by 1. We then show that, once Reductions 8–10 (or the earlier reduction rules) no longer apply, the two trees have a size that is bounded from above by $9k-8$. Additionally, we establish that this bound is (essentially) tight by describing irreducible pairs of trees with TBR distance k that have $9k-9$ leaves.

We anticipate that the new reduction rules will yield new advances for other agreement-forest based distances in phylogenetics, contribute to a deeper understanding of the combinatorics of agreement forests, and facilitate the ongoing advancement of kernelization within phylogenetics. We reflect more on this in the conclusion section at the end of the article, and end the introduction by formally stating the main result of the paper.

Theorem 1.1. *Let T and T' be two unrooted binary phylogenetic trees with leaf set X that cannot be reduced under any of Reductions 1–10. If $d_{\mathrm{TBR}}(T,T') > 2$, then $|X| \leq 9d_{\mathrm{TBR}}(T,T') - 8$.*

2. Preliminaries

2.1. **Notation and terminology.**

Our notation closely follows [20]. Throughout this paper, X denotes a non-empty finite set of *taxa*.

Phylogenetic trees. An *unrooted binary phylogenetic tree* T on X is an undirected tree whose leaves are bijectively labeled with X and whose other vertices all have degree 3. The set X is often referred to as the *leaf set* of T. See Figure 1 for an example of two unrooted binary phylogenetic trees on $X = \{a, b, c, d, e\}$. For simplicity and since all phylogenetic trees in this paper are unrooted and binary, we refer to an unrooted binary phylogenetic trees as a *phylogenetic tree*. Two leaves, say a and b, of T are called a *cherry* $\{a, b\}$ of T if they are adjacent to a common vertex. Moreover, for each $x \in X$, we use p_x to denote the unique neighbor of x in T and refer to p_x as the *parent* of x.

For $X' \subseteq X$, we write $T[X']$ to denote the unique, minimal subtree of T that connects all elements in X'. For brevity we call $T[X']$ the *embedding* of X' in T. For an edge e of T, we say that $T[X']$ *uses* e, if e is an edge of $T[X']$. Furthermore, we refer to the phylogenetic tree on X' obtained from $T[X']$ by suppressing degree-2 vertices as the *restriction of* T *to* X' and we denote this by $T|X'$.

Subtrees and chains. Let T be a phylogenetic tree on X. We say that a subtree of T is *pendant* if it can be detached from T by deleting a single edge. For $n \geq 2$, let $C = (\ell_1, \ell_2 \ldots, \ell_n)$ be a sequence of distinct taxa in X. We call C an n-chain of T if there exists a walk $p_{\ell_1}, p_{\ell_2}, \ldots, p_{\ell_n}$ in T and the vertices in $p_{\ell_2}, p_{\ell_3}, \ldots, p_{\ell_{n-1}}$ are all pairwise distinct. Note that ℓ_1 and ℓ_2 may have a common parent or ℓ_{n-1} and ℓ_n may have a common parent. Furthermore, if $p_{\ell_1} = p_{\ell_2}$ or $p_{\ell_{n-1}} = p_{\ell_n}$ holds, then C is said to be *pendant* in T. To ease reading, we sometimes write C to denote the set $\{\ell_1, \ell_2, \ldots, \ell_n\}$. It will always be clear from the context whether C refers to the associated sequence or set of taxa. If a pendant subtree S (resp. an n-chain C) exists in two phylogenetic trees T and T' on X, we say that S (resp. C) is a *common* subtree (resp. chain) of T and T'.

Tree bisection and reconnection. Let T be a phylogenetic tree on X. Apply the following three-step operation to T:

(1) Delete an edge in T and suppress any resulting degree-2 vertex. Let T_1 and T_2 be the two resulting phylogenetic trees.
(2) If T_1 (resp. T_2) has at least one edge, subdivide an edge in T_1 (resp. T_2) with a new vertex v_1 (resp. v_2) and otherwise set v_1 (resp. v_2) to be the single isolated vertex of T_1 (resp. T_2).
(3) Add a new edge $\{v_1, v_2\}$ to obtain a new phylogenetic tree T' on X.

We say that T' has been obtained from T by a single *tree bisection and reconnection (TBR) operation*. Furthermore, we define the TBR *distance* between two phylogenetic trees T and T' on X, denoted by $d_{\text{TBR}}(T, T')$, to be the minimum number of TBR operations that are required to transform T into T'. To illustrate, the trees T and T' in Figure 2 have a TBR

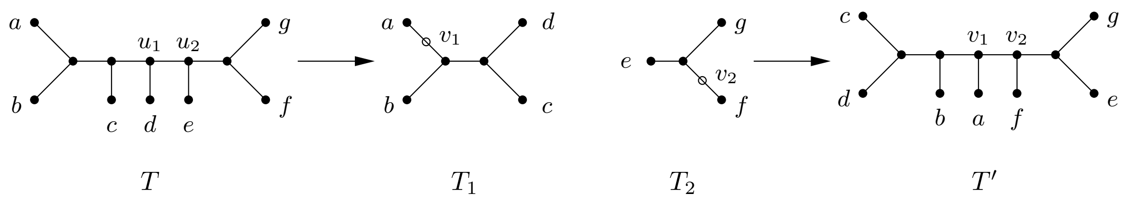

FIGURE 2. A single TBR operation that transforms T into T'. First, up to ignoring the open circle degree-2 vertices, T_1 and T_2 are obtained from T by deleting the edge $\{u_1, u_2\}$ in T. Second, T' is obtained from T_1 and T_2 by subdividing an edge in both trees as indicated by the open circles v_1 and v_2 and adding a new edge $\{v_1, v_2\}$.

distance of 1. It is well known that d_{TBR} is a metric [2]. By building on an earlier result by Hein et al. [15, Theorem 8], Allen and Steel [2] showed that computing the TBR distance is an NP-hard problem.

Agreement forests. Let T and T' be two phylogenetic trees on X. Furthermore, let $F = \{U_0, U_1, U_2, \ldots, U_k\}$ be a partition of X, where each block U_i with $i \in \{0, 1, 2, \ldots, k\}$ is referred to as a *component* of F. We say that F is an *agreement forest* for T and T' if the following conditions hold.

(1) For each $i \in \{0, 1, 2, \ldots, k\}$, we have $T|U_i = T'|U_i$.
(2) For each pair $i, j \in \{0, 1, 2, \ldots, k\}$ with $i \neq j$, we have that $T[U_i]$ and $T[U_j]$ are vertex-disjoint in T, and $T'[U_i]$ and $T'[U_j]$ are vertex-disjoint in T'.

Let $F = \{U_0, U_1, U_2, \ldots, U_k\}$ be an agreement forest for T and T'. The *size* of F is simply its number of components; i.e. $k + 1$. Moreover, an agreement forest with the minimum number of components (over all agreement forests for T and T') is called a *maximum agreement forest (MAF)* for T and T'. The number of components of a maximum agreement forest for T and T' is denoted by $d_{\mathrm{MAF}}(T, T')$. The following theorem is well known.

Theorem 2.1. [2, Theorem 2.13] *Let T and T' be two phylogenetic trees on X. Then*

$$d_{\mathrm{TBR}}(T, T') = d_{\mathrm{MAF}}(T, T') - 1.$$

A maximum agreement forest for the trees T and T' shown in Figure 2, which have TBR distance 1, therefore contains two components. Here $F = \{\{a, b, c, d\}, \{e, f, g\}\}$ is the unique maximum agreement forest for T and T'.

Phylogenetic networks. An *unrooted binary phylogenetic network* N on X is a simple, connected, and undirected graph whose leaves are bijectively labeled with X and whose other vertices all have degree 3. Let E and V be the edge and vertex set of N, respectively. As with phylogenetic trees, we refer to an unrooted binary phylogenetic network simply as a *phylogenetic network*. Furthermore, we define the *reticulation number* of a phylogenetic network N as the number of edges in E that need to be deleted from N to obtain a spanning tree. More formally, we have $r(N) = |E| - (|V| - 1)$. If $r(N) = 0$, then N is simply a phylogenetic tree on X.

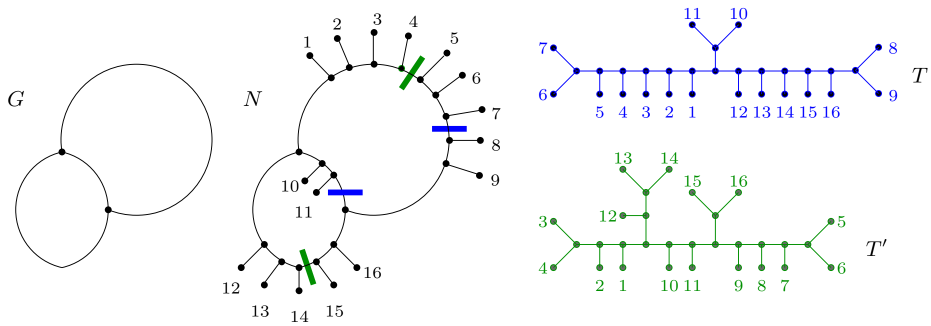

FIGURE 3. The generator G (left) underlying the phylogenetic network N (middle) that displays T and T' (right). An image of T (respectively, T') can be obtained by deleting the $r(N) = d_{\text{TBR}}(T, T') = 2$ blue (respectively, green) breakpoints. The generator underlying N has three sides: a 2-breakpoint side with 9 taxa, 1234|567|89, and two 1-breakpoint sides. Note that due to the common chain $C = (1, 2, 3, 4)$ these trees could be reduced further by Reduction 2.

Let N be a phylogenetic network on X, and let T be a phylogenetic tree on X. We say that N *displays* T if, up to suppressing degree-2 vertices, T can be obtained from N by deleting edges and vertices, in which case, the resulting subgraph of N is an *image* of T in N. Observe that an image of T in N is a subdivision of T. See Figure 3 for an example of the notion of displaying.

Generators. Let k be a positive integer. For $k \geq 2$, a *k-generator* (or short *generator* when k is clear from the context) is a connected cubic multigraph with edge set E and vertex set V such that $k = |E| - (|V| - 1)$. The edges of a generator are called its *sides*. Intuitively, given a phylogenetic network N with $r(N) = k$, we can obtain a k-generator by, repeatedly, deleting all (labeled and unlabeled) leaves and suppressing any resulting degree-2 vertices. We say that the generator obtained in this way *underlies* N. An example of a 2-generator is shown in Figure 3. Now, let G be a k-generator, let $\{u, v\}$ be a side of G, and let Y be a set of leaves. The operation of subdividing $\{u, v\}$ with $|Y|$ new vertices and, for each such new vertex w, adding a new edge $\{w, \ell\}$, where $\ell \in Y$ and Y bijectively labels the new leaves, is referred to as *attaching* Y to $\{u, v\}$ or as *decorating* $\{u, v\}$ *with* Y. Lastly, if at least one new leaf is attached to each loop and to each pair of parallel edges in G, then the resulting graph is a phylogenetic network N with $r(N) = k$. Note that N has no pendant subtree with more than a single leaf.

Hence, we have the following observation.

Observation 2.2. *Let N be a phylogenetic network that has no pendant subtree with at least two leaves, and let G be a generator. Then G underlies N if and only if N can be obtained from G by attaching a (possibly empty) set of leaves to each side of G.*

Unrooted minimum hybridization. In [29], it was shown that computing the TBR distance for a pair of phylogenetic trees T and T' on X is equivalent to computing the minimum number of extra edges required to simultaneously explain T and T'. More precisely, we set

$$h^u(T, T') = \min_N \{r(N)\},$$

where the minimum is taken over all phylogenetic networks N on X that display T and T' (and possibly other phylogenetic trees). The value $h^u(T, T')$ is known as the *(unrooted) hybridization number* of T and T' [29].

The aforementioned equivalence is given in the next theorem that was established in [29, Theorem 3].

Theorem 2.3. *Let T and T' be two phylogenetic trees on X. Then*

$$d_{\mathrm{TBR}}(T, T') = h^u(T, T').$$

This means that $d_{\mathrm{TBR}}(T, T') \leq k$ if and only if there exists a phylogenetic network N with $r(N) \leq k$ that displays both T and T'. If T and T' do not have a common pendant subtree with at least two leaves such an N can be obtained from its underlying generator, which has exactly $3(k-1)$ sides [19, Lemma 1], by attaching taxa to sides. The articles [19, 20] use this fact extensively to derive a bound on the size of the kernelized instance. We will use the same generator-based framework for our results.

Parameterized algorithms. A *parameterized problem* is a problem for which the inputs are of the form (x, k), where k is a non-negative integer, called the *parameter*. A parameterized problem is *fixed-parameter tractable* (FPT) if there exists an algorithm that solves[1] any instance (x, k) in $f(k) \cdot |x|^{O(1)}$ time, where $f(\cdot)$ is a computable function depending only on k. A parameterized problem P has a *kernel* of size $g(k)$, where $g(\cdot)$ is a computable function depending only on k, if there exists a polynomial time algorithm transforming any instance (x, k) into an instance (x', k') of P such that (x, k) is a yes-instance if and only if (x', k') is a yes-instance, and $|x'|, k' \leq g(k)$. Informally, this polynomial-time algorithm usually consists of reduction rules that are applied to an instance (x, k) to transform it into an equivalent but smaller instance (x', k'). If $g(k)$ is a polynomial in k then we call this a *polynomial kernel*; if $g(k) = O(k)$ then it is a *linear kernel*. It is well-known that a parameterized problem is fixed-parameter tractable if and only if it has a (not necessarily polynomial) kernel. For more background information on fixed parameter tractability and kernelization, we refer the reader to standard texts such as [8, 9, 14].

Let T and T' be two phylogenetic trees on X. To compute $d_{\mathrm{TBR}}(T, T')$, we take d_{TBR} as the parameter k and take $|X|$, the number of leaves, as the size of the instance $|x|$. The reduction rules described in the following section produce a linear kernel and run in poly$(|X|)$ time.

[1]Note that the formalism described here actually concerns *decision* (i.e. yes/no) problems, which in the context of the current article is most naturally "Is $d_{\mathrm{TBR}}(T, T') \leq k$?". An FPT algorithm for answering this question can easily be transformed into an algorithm for computing d_{TBR} with similar asymptotic time complexity by increasing k incrementally from 0 until a yes-answer is obtained.

2.2. **Seven reductions to kernelize the TBR distance.**

We start this section by describing the existing seven reductions that have previously been used to establish kernelization results for computing the TBR distance. These existing reductions will be extended to ten reductions in Section 5.

Let T and T' be two phylogenetic trees on X. Each of the existing seven reductions either preserves the TBR distance or reduces it by one as stated and formally shown in [2] for Reductions 1 and 2 and in [20] for Reductions 3–7.

Reduction 1. [2] If T and T' have a maximal common pendant subtree S with at least two leaves, then reduce T and T' to T_r and T'_r, respectively, by replacing S with a single leaf with a new label. This reduction is TBR distance preserving, i.e., $d_{\text{TBR}}(T, T') = d_{\text{TBR}}(T_r, T'_r)$.

Reduction 2. [2] If T and T' have a maximal common n-chain $C = (\ell_1, \ell_2, \ldots, \ell_n)$ with $n \geq 4$, then reduce T and T' to $T_r = T|X \setminus \{\ell_4, \ell_5, \ldots, \ell_n\}$ and $T'_r = T'|X \setminus \{\ell_4, \ell_5, \ldots, \ell_n\}$, respectively. This reduction is TBR distance preserving, i.e., $d_{\text{TBR}}(T, T') = d_{\text{TBR}}(T_r, T'_r)$.

Reduction 3. [20] If T and T' have a common 3-chain $C = (\ell_1, \ell_2, \ell_3)$ such that $\{\ell_1, \ell_2\}$ is a cherry in T and $\{\ell_2, \ell_3\}$ is a cherry in T', then reduce T and T' to $T_r = T|X \setminus C$ and $T'_r = T'|X \setminus C$, respectively. This reduction is TBR distance reducing, i.e., $d_{\text{TBR}}(T, T') - 1 = d_{\text{TBR}}(T_r, T'_r)$.

Reduction 4. [20] If T and T' have a common 3-chain $C = (\ell_1, \ell_2, \ell_3)$ such that $\{\ell_2, \ell_3\}$ is a cherry in T and $\{\ell_3, x\}$ is a cherry in T' with $x \in X \setminus C$, then reduce T and T' to $T_r = T|X \setminus \{x\}$ and $T'_r = T'|X \setminus \{x\}$, respectively. This reduction is TBR distance reducing, i.e., $d_{\text{TBR}}(T, T') - 1 = d_{\text{TBR}}(T_r, T'_r)$.

Reduction 5. [20] If T and T' have two common 2-chains $C_1 = (\ell_1, \ell_2)$ and $C_2 = (\ell_3, \ell_4)$ such that T has cherries $\{\ell_2, x\}$ and $\{\ell_3, \ell_4\}$, and T' has cherries $\{\ell_1, \ell_2\}$ and $\{\ell_4, x\}$ with $x \in X \setminus (C_1 \cup C_2)$, then reduce T and T' to $T_r = T|X \setminus \{x\}$ and $T'_r = T'|X \setminus \{x\}$, respectively. This reduction is TBR distance reducing, i.e., $d_{\text{TBR}}(T, T') - 1 = d_{\text{TBR}}(T_r, T'_r)$.

Reduction 6. [20] If T and T' have two common 3-chains $C_1 = (\ell_1, \ell_2, \ell_3)$ and $C_2 = (\ell_4, \ell_5, \ell_6)$ such that T has cherries $\{\ell_2, \ell_3\}$ and $\{\ell_4, \ell_5\}$, and $(\ell_1, \ell_2, \ldots, \ell_6)$ is a 6-chain of T', then reduce T and T' to $T_r = T|X \setminus \{\ell_4, \ell_5\}$ and $T'_r = T'|X \setminus \{\ell_4, \ell_5\}$, respectively. This reduction is TBR distance preserving, i.e., $d_{\text{TBR}}(T, T') = d_{\text{TBR}}(T_r, T'_r)$.

Reduction 7. [20] If T and T' have common chains $C_1 = (\ell_1, \ell_2, \ell_3)$ and $C_2 = (\ell_4, \ell_5)$ such that T has cherries $\{\ell_2, \ell_3\}$ and $\{\ell_4, \ell_5\}$, and $(\ell_1, \ell_2, \ldots, \ell_5)$ is a 5-chain of T', then reduce T and T' to $T_r = T|X \setminus \{\ell_4\}$ and $T'_r = T'|X \setminus \{\ell_4\}$, respectively. This reduction is TBR distance preserving, i.e., $d_{\text{TBR}}(T, T') = d_{\text{TBR}}(T_r, T'_r)$. An example of Reduction 7 is illustrated in Figure 4.

Reduction 1 is known as *subtree reduction* while Reduction 2 is known as *chain reduction* in the literature. Now, suppose that two phylogenetic trees T_r and T'_r have a common 3-chain $C = (\ell_1, \ell_2, \ell_2)$. We refer to the reverse of Reduction 2 which is the process of obtaining T

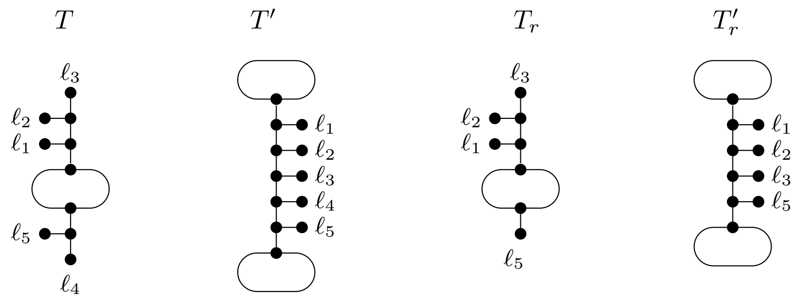

FIGURE 4. An example of Reduction 7. Ovals indicate subtrees.

and T' from T_r and T'_r, respectively, as *extending C to an n-chain* for $n > 3$. We will always explicitly say in which order and to which end of C we add the new leaves $\ell_4, \ell_5, \ldots, \ell_n$.

The following theorem summarizes results established in [2, 19, 20].

Theorem 2.4. *Let S and S' be two phylogenetic trees on X that cannot be reduced by Reduction 1 or 2, and let T and T' be two phylogenetic trees on Y that cannot be reduced by any of Reductions 1–7. If $d_{\mathrm{TBR}}(S, S') \geq 2$, then $|X| \leq 15 d_{\mathrm{TBR}}(S, S') - 9$. If $d_{\mathrm{TBR}}(T, T') \geq 2$, then $|Y| \leq 11 d_{\mathrm{TBR}}(T, T') - 9$.*

Note that each of Reductions 3, 4, and 5 triggers a *parameter reduction*, whereby the TBR distance is reduced by one. In these cases, an element of X is located which definitely comprises a singleton component in some maximum agreement forest, and whose deletion thus lowers the TBR distance by 1. Reductions 1, 2, 6 and 7, on the other hand, preserve the TBR distance. Reduction 6 and 7 work by truncating short chains, i.e. chains which escape Reduction 2, to be even shorter.

The following minor observation is worth noting.

Observation 2.5. *Assume that Reductions 1–7 have been applied to exhaustion. Suppose T and T' have a common chain $C = (b, c, d)$ that is pendant in T'. Then C is not pendant in T.*

Proof. If C was pendant in T then at least one of the subtree reduction or Reduction 3 would be applicable on C, contradicting the assumption that the reduction rules had been applied to exhaustion. □

We end this section by outlining some of the machinery used in [19, 20] to kernelize the TBR distance. This article builds on that machinery and further refines it. Let T and T' be two phylogenetic trees on X that cannot be reduced under Reduction 1 or 2, and let N be a phylogenetic network on X that displays T and T'. Let R and R' be spanning trees of N obtained by greedily extending an image of T (respectively, T') to become a spanning tree, if it is not that already. Since N displays T and T', R and R' exist. Furthermore, let G be the generator that underlies N. Since T and T' are subtree and chain reduced, N does not have a pendant subtree of size at least two. Hence, by Observation 2.2, we can obtain N from G

by attaching leaves to G. Let $S = \{u, w\}$ be a side of G. Let $Y = \{\ell_1, \ell_2, \ldots, \ell_m\}$ be the set of leaves that are attached to S in obtaining N from G. Recall that $m \geq 0$. Then there exists a path

$$u = v_0, v_1, v_2, \ldots, v_m, v_{m+1} = w$$

of vertices in N such that, for each $i \in \{1, 2, \ldots, m\}$, v_i is the unique neighbor of ℓ_i. We refer to this path as the *path associated with* S and denote it by P_S. Importantly, for a path P_S in N that is associated with a side S of G, there is at most one edge in P_S that is not contained in R, and there is at most one (not necessarily distinct) edge in P_S that is not contained in R'. We make this precise in the following definition and say that S *is a b-breakpoint side* relative to R and R', where

(1) $b = 0$ if R and R' both contain all edges of P_S,
(2) $b = 1$ if one element in $\{R, R'\}$ contains all edges of P_S while the other element contains all but one edge of P_S, and
(3) $b = 2$ if each of R and R' contains all but one edge of P_S.

Since R and R' span N, note that S cannot have more than two breakpoints relative to R and R'. Let $S = \{u, w\}$ be a side of G to which t taxa $\{\ell_1, \ell_2, \ldots, \ell_t\}$ get attached in obtaining N from G, and let $P_S = u, p_{\ell_1}, p_{\ell_2}, \ldots, p_{\ell_t}, w$ be the path associated with S. For shorthand, we use $t_1|t_2$ with $t_1 + t_2 = t$, $t_1 \geq 0$ and $t_2 \geq 0$ to refer to S if P_S has a single breakpoint such that one of R and R' does not contain the edge $\{p_{\ell_{t_1}}, p_{\ell_{t_1+1}}\}$. Alternatively, we write $S = \ell_1\ell_2 \cdots \ell_{t_1}|\ell_{t_1+1}\ell_{t_1+2} \cdots \ell_t$. Similarly, if P_S has two breakpoints, we use $t_1|t_2|t_3$ and $S = \ell_1\ell_2 \cdots \ell_{t_1}|\ell_{t_1+1}\ell_{t_1+2} \cdots \ell_{t_1+t_2}|\ell_{t_1+t_2+1}\ell_{t_1+t_2+2} \cdots \ell_t$ with $t_1 + t_2 + t_3 = t$, $t_1 \geq 0$, $t_2 \geq 0$, and $t_3 \geq 0$, where one of R and R' does not contain the edge $\{p_{\ell_{t_1}}, p_{\ell_{t_1+1}}\}$ and the other does not contain the edge $\{p_{\ell_{t_1+t_2}}, p_{\ell_{t_1+t_2+1}}\}$. See Figure 3 for an example illustrating the breakpoint notation. If R and R' both have the same breakpoint (i.e. there exists an edge of P_S that neither R nor R' contains and, in particular $t_2 = 0$), then we write, for example, $t_1||t_3$. Lastly, note that there also may exist a side such that R or R' does not contain the edge $\{u, p_1\}$ or $\{p_t, w\}$ in which case $t_i = 0$ for an element $i \in \{1, 2\}$ if P_S has a single breakpoint and $i \in \{1, 3\}$ if P_S has two breakpoints.

3. Two technical results about short chains

This section presents two technical but powerful theorems that play a crucial part in the upcoming sections. The first, Theorem 3.1, was established in [20, Theorem 5], while the second, Theorem 3.2, is new to this paper.

Let $F = \{U_0, U_1, U_2, \ldots, U_k\}$ be an agreement forest for two phylogenetic trees T and T' on X, and let Y be a subset of X. We say that Y is *preserved* in F if there exists a component U_i in F with $i \in \{0, 1, 2, \ldots, k\}$ such that $Y \subseteq U_i$. Throughout the article we will make heavy use of the following theorem, referred to as the *chain preservation theorem (CPT)*.

Theorem 3.1. *Let T and T' be two phylogenetic trees on X. Let K be an (arbitrary) set of mutually taxa-disjoint chains that are common to T and T'. Then there exists a maximum agreement forest F of T and T' such that*

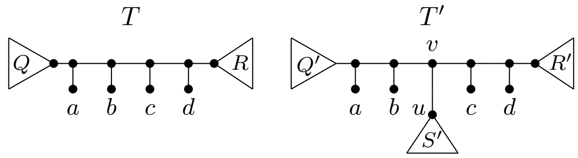

Figure 5. Here (a, b, c, d) is an interrupted 4-chain of T and T'. Triangles indicate subtrees of T and T'. The sets Q, R, Q', R', S' are referred to in the proof of Theorem 3.2, which proves that at least one maximum agreement forest of T, T' does *not* use the edge $\{u, v\}$ in T'. Note that S' must contain at least one leaf, but the leaf set of any of the subtrees Q, R, Q', and R' may be empty, in which case $\{a, b\}$ or $\{c, d\}$ can become cherries.

(1) every n-chain in K with $n \geq 3$ is preserved in F, and
(2) every 2-chain in K that is pendant in at least one of T and T' is preserved in F.

Following on from the last theorem, we say that common n-chains with $n \geq 3$, and common 2-chains that are pendant in at least one of T and T' are *CPT-eligible* chains. In our proofs, CPT-eligible chains will function as 'obstructions' that allow us to reason about the structure of maximum agreement forests.

We now turn to the second technical result. Let $C = (a, b, c, d)$ be a 4-chain. We say that C is an *interrupted 4-chain* of two phylogenetic trees T and T' on X if C is a chain of T and, in T', there exists a walk p_a, p_b, v, p_c, p_d such that p_b, v, and p_c are three pairwise distinct vertices. Furthermore, the edge $e = \{u, v\}$ in T' with $u \notin \{p_b, p_c\}$ is called the *interrupter* of C. Note that v is not necessarily the parent of a leaf in T', and that T and the tree resulting from deleting e in T' and suppressing v have C as a common 4-chain. An example of an interrupted 4-chain is shown in Figure 5.

Theorem 3.2. *Let T and T' be two phylogenetic trees on X, and let $C = (a, b, c, d)$ be an interrupted 4-chain of T and T'. Then there exists a maximum agreement forest F for T and T' such that, for each $U \in F$, $T'[U]$ does not use the interrupter of C in T'.*

Proof. Let $e = \{u, v\}$ be the interrupter in T' of C, and let v be the unique common neighbor of p_b and p_c in T'. Towards a contradiction assume that the result does not hold. Let F^* be a maximum agreement forest for T and T'. Then there exists a component $U \in F^*$ such that $T'[U]$ uses e. Let Q, R be the bipartition of $X \setminus \{a, b, c, d\}$ such that, in T, the path from each leaf in Q to a is shorter than its path to d, and the path from each leaf in R to d is shorter than its path to a. Similarly, let Q', R', S' be the tripartition of $X \setminus \{a, b, c, d\}$ such that, in T', the path from each leaf in Q' to a is shorter than its path to d, the path from each leaf in R' to d is shorter than its path to a, and the path from each leaf in S' to a has the same length than its path to d. This setup is illustrated in Figure 5. We next define five sets that will be useful throughout the proof. Specifically, let $U_Q = U \cap Q$, $U_R = U \cap R$, $U_{Q'} = U \cap Q'$, $U_{R'} = U \cap R'$, and $U_{S'} = U \cap S'$. As $T'[U]$ uses e, note that $U_{S'}$ is non empty. Moreover, since $T|U = T'|U$, it follows that $|U \cap C| \leq 3$. We freely use the previous two properties of $U_{S'}$ and $U \cap C$, respectively, throughout the remainder of the proof. To establish the result, we next consider three cases that each have several subcases. In all (sub)cases we will show that there exists a maximum agreement forest F that does not use e. The exposition of the first case (Case 1 below) is particularly detailed in arguing why F satisfies Condition (2) in the definition of an agreement forest. The arguments for the remaining cases are similar.

Case 1. $U_{Q'} = \emptyset$, and $U_{R'} = \emptyset$
Since $T'[U]$ uses e, we have $1 \leq |U \cap C| \leq 3$. Furthermore, there is no component in $F^* \setminus \{U\}$ that contains a leaf of Q' and a leaf of R'. However, if $U_{S'} \subseteq Q$ or $U_{S'} \subseteq R$, then there can be a component in $F^* \setminus U$ that has a non-empty intersection with C and a non-empty intersection with one of Q and R. There are three subcases to consider for Case 1.

First suppose that $U \cap \{a, b\} \neq \emptyset$ and $U \cap \{c, d\} \neq \emptyset$. If $|U \cap C| = 2$, then $U_{S'} \subseteq Q$ or $U_{S'} \subseteq R$. On the other hand, if $|U \cap C| = 3$ then, because $T[U] = T'[U]$, we have $U_{S'} \subseteq R$ when $\{a, b\} \subset U$, and $U_{S'} \subseteq Q$ when $\{c, d\} \subset U$. Considering $T[U]$, it follows that there exists a leaf $\ell \in C \setminus U$ such that $\{\ell\} \in F^*$. Moreover, p_ℓ is not a vertex of the embedding of any element of $(F^* \setminus \{U\}) \cup \{U_{S'}\}$ in T or T'. Hence, as F^* is an agreement forest for T and T', it now follows that

$$F = (F^* \setminus \{U, \{\ell\}\}) \cup \{U_{S'}, (U \setminus U_{S'}) \cup \{\ell\}\}$$

is such a forest T and T' with $|F| = |F^*|$.

Second suppose that $U \cap \{a, b\} = \emptyset$ and $|U \cap \{c, d\}| = 2$. Then again $U_{S'} \subseteq Q$ or $U_{S'} \subseteq R$. Let U' be the component in F^* that contains b. Note that $|U'| \geq 1$ if $U_{S'} \subseteq R$, and that $|U'| = 1$ if $U_{S'} \subseteq Q$. Moreover if U' contains a leaf in $X \setminus C$, then $U' \setminus C \subseteq Q$ and $U' \setminus C \subseteq Q'$. In turn, this implies that the edge $\{p_b, p_c\}$ in T is not an edge of the embedding of any element of $(F^* \setminus \{U\}) \cup \{U_{S'}\}$ in T and, similarly, the edge $\{p_b, v\}$ in T' is not an edge of the embedding of any element of F^* in T'. Hence, as F^* is an agreement forest for T and T', it now follows that

$$F = (F^* \setminus \{U, U'\}) \cup \{U_{S'}, (U \setminus U_{S'}) \cup U'\}$$

is such a forest for T and T' with $|F| = |F^*|$. An analogous symmetric analysis applies when $|U \cap \{a, b\}| = 2$ and $U \cap \{c, d\} = \emptyset$.

Third suppose that $U \cap \{a, b\} = \emptyset$ and $|U \cap \{c, d\}| = 1$. If $\{c, d\} \cap U = \{c\}$, let U' be the component in F^* that contains b, and if $\{c, d\} \cap U = \{d\}$, let $U' = \{c\}$. In the latter case, note that $U' \in F^*$ because of $T'[U]$. Now assume that $U_Q = \emptyset$ or $U_R = \emptyset$. If $\{c, d\} \cap U = \{c\}$, then similar to the previous case, observe that the edges $\{p_b, p_c\}$ in T and $\{p_b, v\}$ in T' are not contained in the embedding of any element of $(F^* \setminus \{U\}) \cup \{U_{S'}\}$ in T or T'. On the other hand, if $\{c, d\} \cap U = \{d\}$, then p_c is not a vertex of the embedding of any element of $(F^* \setminus \{U\}) \cup \{U_{S'}\}$ in T or T'. As F^* is an agreement forest for T and T', it now follows that

$$F = (F^* \setminus \{U, U'\}) \cup \{U_{S'}, (U \setminus U_{S'}) \cup U'\}$$

is such a forest for T and T' with $|F| = |F^*|$. We may therefore assume that $U_Q \neq \emptyset$ and $U_R \neq \emptyset$, that is $U_{S'} = U_Q \cup U_R$. Then, because of $T[U]$, there are three singletons $\{\ell\}, \{\ell'\}$, and $\{\ell''\}$ in F^* such that $\{\ell, \ell', \ell''\} = C \setminus U$. In other words, there is no component in $F^* \setminus \{U\}$ that contains a leaf in C and a leaf not in C. Furthermore, no edge on the path from p_a to p_d in T and T' is an edge of the embedding of any element of $(F^* \setminus \{U\}) \cup \{U_Q, U_R\}$ in T or T'. Thus

$$F = (F^* \setminus \{U, \{\ell\}, \{\ell'\}, \{\ell''\}\}) \cup \{U_Q, U_R, (U \setminus U_{S'}) \cup \{\ell, \ell', \ell''\}\}$$

is an agreement forest for T and T' with $|F| < |F^*|$. An analogous symmetric analysis applies when $|U \cap \{a,b\}| = 1$ and $U \cap \{c,d\} = \emptyset$.

Case 2. $U_I = \emptyset$ and $U_J \neq \emptyset$ with $\{I,J\} = \{Q',R'\}$
Without loss of generality, we may assume that $U_{Q'} = \emptyset$ and $U_{R'} \neq \emptyset$. There are four subcases to consider for Case 2.

First suppose that $U \cap \{a,b\} \neq \emptyset$ and $U \cap \{c,d\} \neq \emptyset$. Since $T|U = T'|U$, it follows that $U_{S'} = U_Q$ and $U_{R'} = U_R$. Moreover, if $\{a,b\} \subset U$, then $T|U \neq T'|U$, which implies that there exists a leaf $\ell \in \{a,b\}$ such that $\{\ell\} \in F^*$. Hence

$$F = (F^* \setminus \{U, \{\ell\}\}) \cup \{U_{S'}, (U \setminus U_{S'}) \cup \{\ell\}\}$$

is an agreement forest for T and T' with $|F| = |F^*|$.

Second suppose that $U \cap \{a,b\} \neq \emptyset$ and $U \cap \{c,d\} = \emptyset$. Then clearly $\{c\}, \{d\} \in F^*$. Since $U \cap C \neq \emptyset$, there exists no component in $F^* \setminus \{U\}$ that contains a leaf of Q and a leaf of R. Moreover, if $U \cap C \in \{\{a\}, \{a,b\}\}$, then no component in $F^* \setminus \{U\}$ contains a leaf in C and a leaf in $X \setminus C$. Hence,

$$F = (F^* \setminus \{U, \{c\}, \{d\}\}) \cup \{U_{S'}, U_{R'}, (U \cap C) \cup \{c,d\}\}$$

is an agreement forest for T and T' with $|F| = |F^*|$. For the remainder of this subcase, assume that $U \cap \{a,b\} = \{b\}$. If $U_Q \neq \emptyset$ then $\{a\} \in F^*$ and F is again an agreement forest for T and T'. Lastly, if $U_Q = \emptyset$, let U' be the component in $F^* \setminus \{U\}$ such that $a \in U'$. As $T'[U]$ and $T'[U']$ are vertex-disjoint, we have $U' \setminus \{a\} \subseteq Q'$. It is now straightforward to check that F is an agreement forest for T and T'.

Third suppose that $U \cap \{a,b\} = \emptyset$ and $U \cap \{c,d\} \neq \emptyset$. If $U \cap \{c,d\} = \{c,d\}$, then $U_{S'} = U_Q$ and $U_{R'} = U_R$. It follows that $\{a\}, \{b\} \in F^*$. On the other hand, if $U \cap \{c,d\} = \{c\}$ (resp. $U \cap \{c,d\} = \{d\}$), then $\{d\} \in F^*$ (resp. $\{c\} \in F^*$). Hence,

$$F = (F^* \setminus \{U, \{\ell\}\}) \cup \{U_{S'}, (U \setminus U_{S'}) \cup \{\ell\}\}$$

is an agreement forest for T and T' with $|F| = |F^*|$ and where $\ell \in \{b,c,d\}$ depending on which of the three leaves is a singleton in F^*.

Fourth suppose that $U \cap C = \emptyset$. Clearly, $\{c\}, \{d\} \in F^*$. If there exists no $U' \in F^*$ such that $U' \cap (Q \cup \{a,b\}) \neq \emptyset$ and $U' \cap R \neq \emptyset$, then

$$F = (F^* \setminus \{U, \{c\}, \{d\}\}) \cup \{U_{S'}, U \setminus U_{S'}, \{c,d\}\}$$

is an agreement forest for T and T' with $|F| = |F^*|$. Hence, we may assume that U' exists. Furthermore, assume first that $U' \neq U$. Since $T|U' = T'|U'$, one of the following properties holds depending on which of a and b is contained in U'.

(1) If $U' \cap \{a,b\} = \emptyset$, then $\{a\}, \{b\} \in F^*$ because $T[U']$ uses $\{p_a, p_b\}$.
(2) If $U' \cap \{a,b\} = \{a\}$, then $\{b\} \in F^*$ because $T[U']$ uses $\{p_a, p_b\}$.
(3) If $U' \cap \{a,b\} = \{b\}$, then $\{a\} \in F^*$ because $T'[U]$ and $T'[U']$ are vertex-disjoint.
(4) If $U' \cap \{a,b\} = \{a,b\}$, then $U' \cap Q = \emptyset$ because $T'[U]$ and $T'[U']$ are vertex-disjoint.

It now follows that

$$F = (F^* \setminus \{U, U', \{a\}, \{b\}, \{c\}, \{d\}\}) \cup \{U_{S'}, U \setminus U_{S'}, C, U' \cap Q, U' \cap R\}$$

is an agreement forest for T and T' with $|F| < |F^*|$ if Property (1) applies,

$$F = (F^* \setminus \{U, U', \{\ell\}, \{c\}, \{d\}\}) \cup \{U_{S'}, U \setminus U_{S'}, C, U' \cap Q, U' \cap R\}$$

is an agreement forest for T and T' with $|F| \leq |F^*|$ and $\ell = b$ (resp. $\ell = a$) if Property (2) (resp. Property (3)) applies, and

$$F = (F^* \setminus \{U, U', \{c\}, \{d\}\}) \cup \{U_{S'}, U \setminus U_{S'}, C, U' \cap R\}$$

is an agreement forest for T and T' with $|F| = |F^*|$ if Property (4) applies. Now assume that $U' = U$. Clearly $\{a\}, \{b\} \in F^*$. Consider the bipartition $U_{S'}, U_{R'}$ of U. Let $U_{S'}^Q = U_{S'} \cap Q$, $U_{S'}^R = U_{S'} \cap R$, $U_{R'}^Q = U_{R'} \cap Q$, and $U_{R'}^R = U_{R'} \cap R$. It now follows that

$$F = (F^* \setminus \{U, \{a\}, \{b\}, \{c\}, \{d\}\}) \cup \{U_{S'}^Q, U_{S'}^R, U_{R'}^Q, U_{R'}^R, C\}$$

is an agreement forest for T and T'. In particular, since $T|U = T'|U$, at least one element in $\{U_{S'}^Q, U_{S'}^R, U_{R'}^Q, U_{R'}^R\}$ is the empty set and, so, $|F| < |F^*|$.

Case 3. $U_{Q'} \neq \emptyset$ and $U_{R'} \neq \emptyset$
Since $T|U = T'|U$, we have that $U \cap \{a, b\} = \emptyset$ or $U \cap \{c, d\} = \emptyset$. Hence $|U \cap C| \leq 2$. There are three subcases to consider for Case 3.

First suppose that $|U \cap C| = 2$. Then there exist two distinct leaves $\ell, \ell' \in C$ such that $\{\ell\}, \{\ell'\} \in F^*$. If $U \cap C = \{a, b\}$ (resp. $U \cap C = \{c, d\}$), then $U_{S'} \cup U_{R'} \subseteq R$ and $U_{Q'} \subseteq Q$ (resp. $U_{S'} \cup U_{Q'} \subseteq Q$ and $U_{R'} \subseteq R$). Hence,

$$F = (F^* \setminus \{U, \{\ell\}, \{\ell'\}\}) \cup \{U_{S'}, (U \setminus U_{S'}) \cup \{\ell, \ell'\}\}$$

is an agreement forest for T and T' with $|F| < |F^*|$.

Second suppose that $|U \cap C| = 1$. Then there exist three distinct leaves $\ell, \ell', \ell'' \in C$ such that $\{\ell\}, \{\ell'\}, \{\ell''\} \in F^*$. In turn, because there is no component in $F^* \setminus U$ that contains a leaf in Q and a leaf in R, this implies that, except for possibly U, no other component in F^* uses any of the three edges $\{p_a, p_b\}$, $\{p_b, p_c\}$, and $\{p_c, p_d\}$. Lastly, since $U \cap C \neq \emptyset$, $U_{S'}$ is either contained in Q or R. It now follows that

$$F = (F^* \setminus \{U, \{\ell\}, \{\ell'\}, \{\ell''\}\}) \cup \{U_{Q'}, U_{R'}, U_{S'}, C\}$$

is an agreement forest for T and T' with $|F| = |F^*|$.

Third suppose that $|U \cap C| = 0$. Then each leaf in C is a singleton in F^*. If there exists no component $U' \in F^*$ such that $T[U']$ uses an edge $\{p_\ell, p_{\ell'}\}$ for two distinct $\ell, \ell' \in C$, then

$$F = (F^* \setminus \{U, \{a\}, \{b\}, \{c\}, \{d\}\}) \cup \{U_{Q'}, U_{R'}, U_{S'}, C\}$$

is an agreement forest for T and T' with $|F| < |F^*|$. Otherwise, if U' exists, then U' is the unique such component since $U' \cap Q \neq \emptyset$ and $U' \cap R \neq \emptyset$. Hence, assuming that $U' \neq U$,

$$F = (F^* \setminus \{U, U', \{a\}, \{b\}, \{c\}, \{d\}\}) \cup \{U_{Q'}, U_{R'}, U_{S'}, U' \cap Q, U' \cap R, C\}$$

is an agreement forest for T and T' with $|F| = |F^*|$. Lastly, if $U' = U$, consider the three sets $U_{Q'}$, $U_{R'}$, and $U_{S'}$. Since $T|U = T'|U$, at most one of these three sets, say $U_{S'}$, has a non-empty intersection with Q and a non-empty intersection with R. Then

$$F = (F^* \setminus \{U, \{a\}, \{b\}, \{c\}, \{d\}\}) \cup \{U_{Q'}, U_{R'}, U_{S'} \cap Q, U_{S'} \cap R, C\}$$

is an agreement forest for T and T' with $|F| = |F^*|$. An analogous argument holds if $U_{Q'}$ or $U_{R'}$ has a non-empty intersection with Q and a non-empty intersection with R. □

4. Main result and a bird's-eye view of the main arguments

In this section, we state the main result of this paper and give an overview of our approach to establish it. The following lemma summarizes the situation after Reductions 1–7 have been applied to exhaustion and is the foundation of Theorem 2.4.

Lemma 4.1. *Let T and T' be two phylogenetic trees on X that cannot be reduced under Reductions 1–7. Let G be the generator underlying a phylogenetic network N on X such that $r(N) = d_{\mathrm{TBR}}(T, T')$. Then, in obtaining N from G, the following statements hold.*

- *(a) At most four taxa can be attached to each side of G.*
- *(b) At most three taxa can be attached to each 0-breakpoint side of G.*
- *(c) At most four taxa can be attached to each 1-breakpoint side of G and only sides of the form $1|3$ and $2|2$ can achieve this upper bound.*
- *(d) At most four taxa can be attached to each 2-breakpoint side of G and only sides of the form $2|1|1$ can achieve this upper bound.*

Proof. In [20, Lemma 7], the authors showed that each 0-breakpoint side of G has at most three taxa and that each other side of G has at most four taxa. Consider a 1-breakpoint side S of G. Suppose that four leaves get attached to S in obtaining N from G. If S is a $0|4$ side, then T and T' have a common 4-chain, contradicting that Reduction 2 has been applied to exhaustion. Hence S is either a $1|3$ or $2|2$ side. Next, consider a 2-breakpoint side S of G. Again, suppose that four leaves get attached to S in obtaining N from G. If S is a $0||4$, $1||3$, or $2||2$ side on which the breakpoints of T and T' coincide, then T and T' have a common subtree with at least two leaves, contradicting that Reduction 1 has been applied to exhaustion. Otherwise, if the breakpoints of T and T' do not coincide and S is a $0|2|2$, $0|1|3$, $0|3|1$, or $0|4|0$ side, then it is straightforward to check that T and T' would have been reduced by Reduction 1, 4, 3 or 2 respectively, again a contradiction. It now follows that S is a $2|1|1$ side. □

Let G be a k-generator as described in Lemma 4.1. By [19, Lemma 1], G has $3(k-1)$ sides, and there are $2k$ breakpoints to divide across these sides. Intuitively, after attaching the elements in X to sides of G in order to obtain N, we delete k edges in N to obtain a subdivision of T and k edges to obtain a subdivision of T'. Instead of deleting edges in N, we think of this as placing breakpoints on the sides of G. Now, with a view towards obtaining an upper bound on the number of leaves that can be attached to any k-generator relative to two phylogenetic trees that are fully reduced under Reductions 1–7, we have $2k$ 1-breakpoint sides with four taxa each, and $(k-3)$ 0-breakpoint sides with three taxa each; this is the origin of the $11k-9$ kernel in [20]. The main bottleneck in achieving a kernel that is smaller than $11k-9$ are sides with four taxa and, in particular, those that only have one breakpoint. This explains the heavy emphasis on $1|3$, $2|2$ and $2|1|1$ sides in the rest of the article.

The high-level idea to achieve a kernel for d_{TBR} that is smaller than $11k-9$ is as follows. In Section 5, we present new Reductions 8, 9, and 10.

(1) A $1|3$ side triggers Reduction 8 that, as long as a 'secondary' common 3-chain is available, reduces the number of taxa by one. The '3' part in a $1|3$ side can itself function as a secondary chain for another $1|3$ side, so this reduction rule eliminates all but at most one $1|3$ side.
(2) A $2|2$ side that (informally) has relatively many taxa on the adjacent sides can be reduced by Reduction 9, which essentially first transforms such a side into a $1|3$ side before executing Reduction 8 if another $1|3$ side (and therefore a 'secondary' common 3-chain) is available.
(3) A $2|1|1$ side that (informally) has relatively many taxa on the adjacent sides triggers the parameter-reducing Reduction 10.

After applying Reductions 1–10 to exhaustion, all sides with four taxa (apart from possibly one single exception) do *not* have many taxa on the adjacent sides. This has the consequence that, once these sparse adjacent sides are taken into account, 4-taxa sides contribute (on average) significantly fewer than four taxa per side. With some careful counting this leads to an improved kernel of size $9k-8$.

We finish this section by noting that the portfolio of reduction rules should always be executed in the order $1, 2, \ldots, 10$. Every time a reduction rule executes, the sequence should be restarted from Reduction 1. The fact that in all cases the number of taxa (and sometimes the TBR distance) is reduced by at least one, and the fact that the reduction rules themselves can be executed in polynomial-time, ensures polynomial-time execution overall.

5. Three new reduction rules

5.1. **Reduction 8: A reduction rule to reduce a $1|3$ side if there is a spare common 3-chain available.**

The reduction rule that we describe first is designed to target the structures of two phylogenetic trees that are induced by a $1|3$ side $S = a|bcd$.

Reduction 8. Let T and T' be two phylogenetic trees on X that cannot be reduced under any of Reductions 1–7. Suppose that T and T' have two leaf-disjoint common 3-chains $C = (b, c, d)$ and $D = (e, f, g)$ such that C is pendant in T' with cherry $\{b, c\}$ and C is not pendant in T, and there exists a taxon a such that (a, b, c, d) is a chain of T and not a chain of T'. First obtain T_1 from T by extending D to the 4-chain (e, f, g, g') such that $g' \notin X$, deleting the edge $\{p_a, p_b\}$, suppressing p_a and p_b, subdividing the edge $\{p_f, p_g\}$ with a new vertex v, and adding the edge $\{u, v\}$, where u is a new vertex that subdivides an arbitrary edge of the component that does not contain e such that (b, c, d) is a pendant 3-chain in T_1.

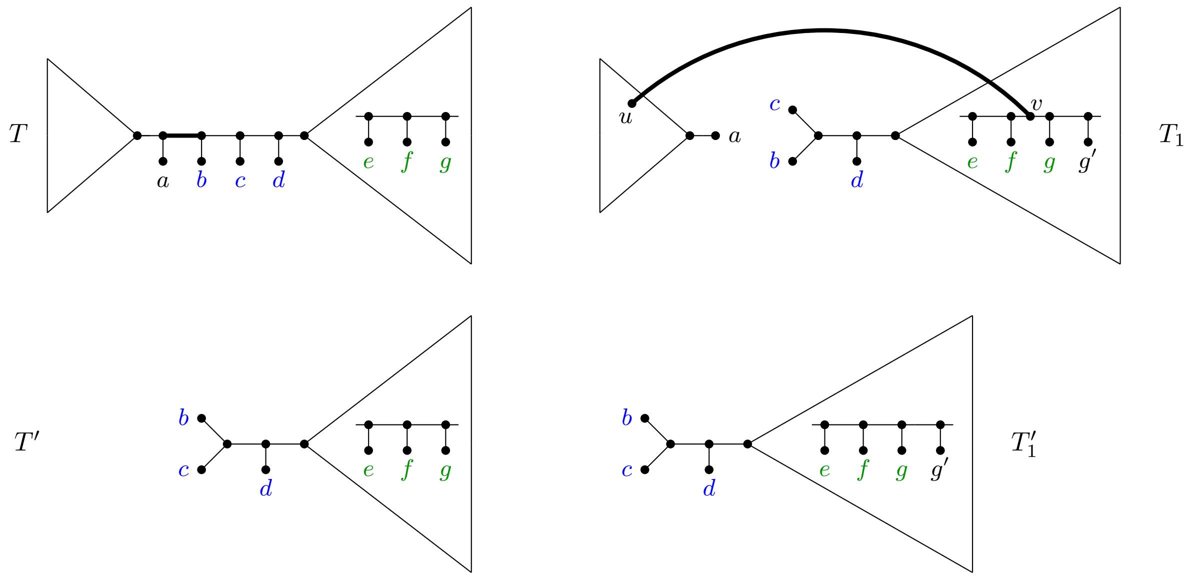

FIGURE 6. The first part of Reduction 8 that reduces a common 3-chain $\{b, c, d\}$ (indicated in blue) of a $1|3$ side $a|bcd$, as long as a secondary common 3-chain $\{e, f, g\}$ (indicated in green) is available. Extending the chain $\{e, f, g\}$ to $\{e, f, g, g'\}$ and then swapping the bold edge in T with the bold edge in T_1 preserves d_{TBR} and creates a common pendant subtree $\{b, c, d\}$ which can be reduced under Reduction 1; the application of Reduction 1 is not shown here. Observe that deleting the edge $\{p_a, p_b\}$ in T disconnects T into two smaller trees and, in the example above, (b, c, d) and (e, f, g) are leaves of the same smaller tree. In general, this does not need to be the case: (b, c, d) and (e, f, g) can be in different subtrees.

We obtain T_1' from T' by extending D to the 4-chain (e, f, g, g'). Then T_1 and T_1' are two phylogenetic trees on $X \cup \{g'\}$. Figure 6 illustrates how T_1 and T_1' are obtained from T and T', respectively. Second, obtain T_r and T_r' from T_1 and T_1', respectively, by an application of Reduction 1 to the common pendant subtree of T_1 and T_1' with leaf set $\{b, c, d\}$. This concludes the definition of Reduction 8.

In what follows, we will call D the *secondary common 3-chain* when executing Reduction 8. Lastly, if T_r and T_r' are obtained from T and T' as described in Reduction 8, we say that *Reduction 8 is applied to* $C = (b, c, d)$ *and* $D = (e, f, g)$, where C and D are as defined in Reduction 8.

The next lemma shows that an application of Reduction 8 preserves the TBR distance and reduces the number of taxa by one. Furthermore, this reduction can be executed in polynomial time by trying all possible candidates for the taxa $\{a, b, c, d, e, f, g\}$ as defined in Reduction 8.

Lemma 5.1. *Let T and T' be two phylogenetic trees on X that cannot be reduced under any of Reductions 1–7. Let T_r and T_r' be two phylogenetic trees on X' that are obtained from T*

and T', respectively, by a single application of Reduction 8. Then $d_{\mathrm{TBR}}(T,T') = d_{\mathrm{TBR}}(T_r,T'_r)$ and $|X'| = |X| - 1$.

Proof. We establish the lemma using the same notation as in the description of Reduction 8. In particular, we denote by T_1 and T'_1 the two trees obtained from T and T', respectively, by the first part of Reduction 8 (i.e. the two trees before Reduction 1 is applied to the common pendant subtree with leaf set $\{b,c,d\}$). Since T and T' cannot be reduced under any of Reductions 1–7, neither C nor D is the leaf set of a common pendant subtree of T and T'. By Observation 2.5, recall that D may be pendant in one of T and T'. If $\{f,g\}$ is a cherry in one of T and T', then interchange the roles of e and g. Hence, without loss of generality, we assume that $\{f,g\}$ is not a cherry in either T or T'. Now, let S and S' be the two phylogenetic trees obtained from T and T', respectively, by extending D to the 4-chain (e,f,g,g'). It follows from the statement of Reduction 2 that $d_{\mathrm{TBR}}(T,T') = d_{\mathrm{TBR}}(S,S')$.

We next show that $d_{\mathrm{TBR}}(S,S') = d_{\mathrm{TBR}}(T_1,T'_1)$ from which the lemma then almost follows immediately. First, let $F = \{U_0,U_1,U_2,\ldots,U_k\}$ be a maximum agreement forest for S and S'. By CPT, we may assume that $C \subseteq U_i$ for some $i \in \{0,1,2,\ldots,k\}$. Let S_1 and S_2 be the two phylogenetic trees obtained from S by deleting the edge $e_1 = \{p_a,p_b\}$ such that S_1 does not contain b. Since C is pendant in S', it follows that U_i does not contain a taxon of S_1 (because otherwise $\{b,c\}$ would be a cherry in $S'|U_i$, but not in $S|U_i$, contradicting the requirement that $S|U_i = S'|U_i$). This in turn implies that e_1 is not used by any embedding $S[U_j]$ with $U_j \in F$, and hence no $U_j \in F$ contains taxa from both S_1 and S_2. Noting that the deletion of the edge $\{u,v\}$ in T_1 splits T_1 into two subtrees with leaf sets S_1 and S_2, it follows that none of the embeddings $T_1[U_j]$, for $U_j \in F$, use the edge $\{u,v\}$. As a consequence, for each $U_j \in F$, we have $S|U_j = T_1|U_j$. The embeddings $T_1[U_j]$ are mutually vertex-disjoint, because the embeddings $S[U_j]$ are mutually disjoint: the embeddings $T_1[U_j]$ only differ from their $S[U_j]$ counterparts by the possible inclusion or suppression of some degree-2 vertices, which cannot compromise disjointness. We conclude that F is an agreement forest for T_1 and T'_1. Second, let $F = \{U_0,U_1,U_2,\ldots,U_k\}$ be a maximum agreement forest for T_1 and T'_1. By Theorem 3.2, we may assume that there exists no component $U_i \in F$ with $i \in \{0,1,2,\ldots,k\}$ such that $T_1[U_i]$ uses the interrupter of (e,f,g,g'). By a symmetrical argument to the above, we then have that the $S[U_j]$ are mutually vertex-disjoint, and that for each $U_j \in F$, $S|U_j = T_1|U_j$. Hence F is an agreement forest for S and S'. Combining both directions establishes that $d_{\mathrm{TBR}}(S,S') = d_{\mathrm{TBR}}(T_1,T'_1)$. Moreover, by construction, $\{b,c,d\}$ is the leaf set of a common pendant subtree of T_1 and T'_1, and, by the statement of Reduction 1, reducing this common pendant subtree to a single new leaf preserves the TBR distance. Hence, letting T_r and T'_r be the two phylogenetic trees obtained from T_1 and T'_1, respectively, by applying Reduction 1 to $\{b,c,d\}$ and noting that $|X'| = |X| + 1 - 2 = |X| - 1$ establishes the lemma. □

Reduction 8 also leads to the following observation, which we will need later.

Observation 5.2. *Let N be a phylogenetic network on X that displays two phylogenetic trees T and T' on X that cannot be reduced under any of Reductions 1–8. Let G be the generator that underlies N. Then G has at most <u>one</u> $1|3$ side.*

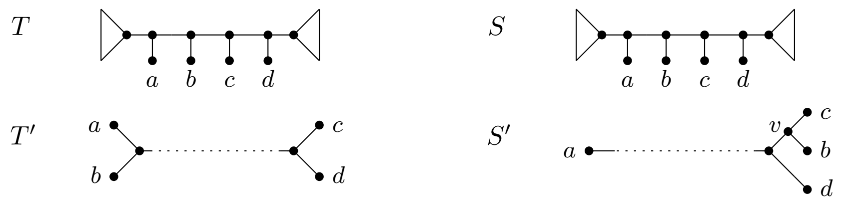

FIGURE 7. A generic example of Operation P, where each triangle indicates a subtrees with at least one leaf and the dotted lines indicate paths in T' and S'.

Proof. Suppose that there are two distinct such sides, $a|bcd$ and $.|efg$ where "." denotes a single taxon. Clearly, $\{b,c,d\} \cap \{e,f,g\} = \emptyset$ because the taxa are from distinct sides of G. Then $C = \{b,c,d\}$ and $D = \{e,f,g\}$ are two common 3-chains of T and T' that satisfy the three properties described in the definition of Reduction 8. Hence, T and T' can be further reduced under Reduction 8, a contradiction. □

5.2. Reduction 9: A reduction rule that triggers Reduction 8 by transforming certain $2|2$ sides into $1|3$ sides.

The next reduction rule targets the structures of two phylogenetic trees that are induced by a $2|2$ side $S = ab|cd$. We start by introducing an operation that does not reduce the number of leaves in the trees. Instead, this operation transforms certain $2|2$ sides of a generator into $1|3$ sides and is a *p*recursor (hence the name P) to Reduction 9 that is described towards the end of this subsection.

Operation P. Let T and T' be two phylogenetic trees on X that cannot be reduced under any of Reductions 1–8. Suppose that T has a non-pendant chain (a,b,c,d), T' has cherries $\{a,b\}$ and $\{c,d\}$, and there exists a maximum agreement forest F for T and T' such that $\{a,b\}$ and $\{c,d\}$ are each preserved in F, but $\{a,b,c,d\}$ is not preserved in F. Then let $S = T$, and let S' be the tree obtained from T' by deleting b, suppressing p_b, subdividing the edge incident with c with a new vertex v, and adding the edge $\{v,b\}$. An example of Operation P is illustrated in Figure 7.

If $\{a,b,c,d\}$ satisfies all properties in the description of Operation P, we say that $\{a,b,c,d\}$ is *eligible for Operation P*. Moreover, if S and S' are obtained from T and T' as described above, we say that *Operation P is applied to* $\{a,b,c,d\}$.

Lemma 5.3. *Let T and T' be two phylogenetic trees on X that cannot be reduced under any of Reductions 1–8. Furthermore, let S and S' be two phylogenetic trees obtained from T and T', respectively, by applying Operation P to $\{a,b,c,d\} \subseteq X$. Then $d_{\mathrm{TBR}}(T,T') = d_{\mathrm{TBR}}(S,S')$. Moreover, (b,c,d) is a common chain of S and S', and S' has cherry $\{b,c\}$.*

Proof. We establish the lemma using the same notation as in the definition of Operation P. First, let $F = \{U_0, U_1, U_2, \ldots, U_k\}$ be a maximum agreement forest for T and T' such that $\{a,b\}$ and $\{c,d\}$ are each preserved in F, but $\{a,b,c,d\}$ is not preserved in F. Let U_j be the component in F such that $\{a,b\} \subseteq U_j$ and, similarly, let $U_{j'}$ be the component in F such that $\{c,d\} \subseteq U_{j'}$. Since the embeddings in $\{T[U_i] : i \in \{0,1,2,\ldots,k\}\}$ are pairwise vertex-disjoint in T, and $\{a,b,c,d\}$ is not preserved in F, observe that no subtree $T[U_i]$ uses the edge $\{p_b, p_c\}$. This implies that the embeddings in

$$\{S[U_i] : i \in \{0,1,2,\ldots,k\} \setminus \{j,j'\}\} \cup \{S[U_j \setminus \{b\}], S[U_{j'} \cup \{b\}]\}$$

are vertex-disjoint in S. Since F is an agreement forest for T and T', it now follows that

$$(F \setminus \{U_j, U_{j'}\}) \cup \{U_j \setminus \{b\}, U_{j'} \cup \{b\}\}$$

is an agreement forest for S and S' that has the same size as F. Hence $d_{\text{TBR}}(S,S') \leq d_{\text{TBR}}(T,T')$. Second, let $F = \{U_0, U_1, U_2, \ldots, U_k\}$ be a maximum agreement forest for S and S'. By CPT, we may assume that $\{b,c,d\} \subseteq U_j$ for some $j \in \{0,1,2,\ldots,k\}$. Since $S|U_j = S'|U_j$ and the subtrees in $\{S[U_i] : i \in \{0,1,2,\ldots,k\}\}$ are pairwise vertex-disjoint, it follows that the edge $\{p_a, p_b\}$ is not used by $S[U_i]$ for any $i \in \{0,1,\ldots,k\}$. Now let $U_{j'}$ be the component in $F \setminus \{U_j\}$ such that $a \in U_{j'}$. Then, as F is an agreement forest for S and S', and applying an argument that is similar to that used to establish the first inequality,

$$(F \setminus \{U_j, U_{j'}\}) \cup \{U_j \setminus \{b\}, U_{j'} \cup \{b\}\}$$

is an agreement forest for T and T' that has the same size as F. Thus $d_{\text{TBR}}(S,S') \geq d_{\text{TBR}}(T,T')$. Combining both cases establishes that $d_{\text{TBR}}(S,S') = d_{\text{TBR}}(T,T')$. Moreover, by construction of S and S' it follows immediately that (b,c,d) is a common chain of S and S', and S' has cherry $\{b,c\}$. □

Let T and T be two phylogenetic trees on X. Following on from the description of Operation P, we present an explicit, polynomial-time algorithm—called Algorithm 1—in Section 7.1 for testing whether or not, given a subset $\{a,b,c,d\}$ of X, there exists a maximum agreement forest F for T and T' such that $\{a,b\}$ and $\{c,d\}$ are each preserved in F, but $\{a,b,c,d\}$ is not preserved in F. Although the algorithm does not necessarily catch *all* situations when F exists, it is enough for our purposes. The high-level idea is that, as soon as a side $ab|cd$ has 'many taxa on its surrounding sides', then T and T' will contain easily detectable structures that constitute a certificate for the existence of F and Algorithm 1 will find them. The correctness of Algorithm 1 is closely linked to the proofs of Lemmas 6.2 and 6.3 in Section 6 and for this reason Algorithm 1 is best studied after reading those proofs.

The following corollary to Lemma 5.3 is useful later.

Corollary 5.4. *Let T and T' be two phylogenetic trees on X that cannot be reduced under any of Reductions 1–8, and let S and S' be the two phylogenetic trees on X that are obtained from T and T', respectively, by applying Operation P to $\{a,b,c,d\} \subseteq X$. Let N be a phylogenetic network on X that displays T and T'. If the generator G that underlies N has a $2|2$ side $S = ab|cd$, then N also displays S and S'.*

Proof. Using the same notation as in the definition of Operation P, observe that the breakpoint on S is relative to T'. To see that N also displays S and S', we view S as the $1|3$ side $a|bcd$, where the breakpoint is now relative to S'. □

Lemma 5.3 and Corollary 5.4 are the theoretical foundation for Operation P. Once Operation P is applied, Reduction 8 may be triggered if another $1|3$ side is available. This can happen in two slightly different ways which we describe next as Reduction 9.1 and 9.2.

Reduction 9.1 is tried first and, if it fails, Reduction 9.2 is tried. Essentially Reduction 9.1 converts one $2|2$ side into a $1|3$ side and Reduction 9.2 converts two $2|2$ sides into $1|3$ sides. In both cases, the new $1|3$ sides trigger Reduction 8. Let T and T' be two phylogenetic trees on X that cannot be reduced under any of Reductions 1–8. Then Reductions 9.1 and 9.2 are defined as follows.

Reduction 9.1. Suppose that there exists $\{a, b, c, d\} \subseteq X$ such that $C = (b, c, d)$ is a common chain of T and T', C is pendant in T' with cherry $\{b, c\}$ and not pendant in T, and (a, b, c, d) is a chain of T and not a chain of T'. Suppose furthermore that there exists $\{a', b', c', d'\} \subseteq X$ which is eligible for Operation P, where $\{a, b, c, d\} \cap \{a', b', c', d'\} = \emptyset$. Then an application of Reduction 9.1 to T and T' consists of an application of Operation P to $\{a', b', c', d'\}$, thereby creating two phylogenetic trees with a common 3-chain (b', c', d'), and a subsequent application of Reduction 8 to C and the newly created secondary common 3-chain $D = (b', c', d')$.

Reduction 9.2. Suppose that there exist two disjoint subsets $\{a', b', c', d'\}$ and $\{a'', b'', c'', d''\}$ of X, such that both are eligible for Operation P. Then, an application of Reduction 9.2 to T and T' consists of an application of Operation P to $\{a', b', c', d'\}$ followed by an application of the same operation to $\{a'', b'', c'', d''\}$ if it is still eligible.[2]

It is important to note that Reduction 9.2 is an 'all-or-nothing' reduction, i.e. it either executes fully or not at all. Specifically, it does not execute the first application of Operation P but not the second. As Algorithm 1 (see Section 7.1) runs in polynomial time, it follows that Reductions 9.1 and 9.2 can be executed in polynomial time by trying all possible candidates for the taxa $\{a, b, c, d, a', b', c', d'\}$ and $\{a', b', c', d', a'', b'', c'', d''\}$, respectively. Lastly, since we do not always need to distinguish between Reduction 9.1 and Reduction 9.2, we refer to an application of one of the two reductions as *Reduction 9*.

5.3. **Reduction 10: A reduction rule to reduce certain $2|1|1$ sides.**

The last new reduction rule targets the structures of two phylogenetic trees that are induced by a $2|1|1$ side $S = ab|c|d$. Reduction 10 is much more straightforward than Reductions 8 and 9.

Reduction 10. Let T and T' be two phylogenetic trees on X that cannot be reduced under any of Reductions 1–9. If T has two cherries $\{a, b\}$ and $\{c, d\}$, T' has the 3-chain (a, b, c) such that $\{b, c\}$ is a cherry, and there exists a maximum agreement forest F for T and T' such that $\{c\} \in F$, then reduce T and T' to $T_r = T|X \setminus \{c\}$ and $T'_r = T'|X \setminus \{c\}$, respectively.

If $\{a, b, c, d\}$ satisfies all properties in the description of Reduction 10, we say that $\{a, b, c, d\}$ is *eligible for Reduction 10*. Moreover, if T_r and T'_r are obtained from T and T' as described above, we say that *Reduction 10 is applied to* $\{a, b, c, d\}$.

[2]In our analysis later in the article, we apply Reduction 9.2 in a situation where $\{a'', b'', c'', d''\}$ is definitely still eligible for transformation after $\{a', b', c', d'\}$ has been transformed, and Algorithm 1 will verify this. Hence, it is not necessary to engage with the intricate combinatorics of the way one application of Operation P could potentially influence another.

The next lemma shows that Reduction 10 is parameter reducing. Its proof is straightforward and omitted.

Lemma 5.5. *Let T and T' be two phylogenetic trees on X that cannot be reduced under any of Reductions 1–9. Let T_r and T_r' be two phylogenetic trees obtained from T and T', respectively, by a single application of Reduction 10. Then $d_{\text{TBR}}(T_r, T_r') = d_{\text{TBR}}(T, T') - 1$.*

It remains to establish that Reduction 10 can be executed in polynomial time. In Section 7.2, we present an explicit, polynomial-time algorithm—called Algorithm 2—for testing whether there exists a maximum agreement forest F for T and T' such that $\{c\} \in F$. (Similar to Algorithm 1, the correctness of Algorithm 2 is closely linked to proofs in Section 6, this time Lemmas 6.4, 6.5 and 6.6. For this reason Algorithm 2 is best studied after reading those proofs.) As Algorithm 2 runs in polynomial time, it follows that Reduction 10 can be executed in polynomial time by trying all possible candidates for the taxa $\{a, b, c, d\}$ as defined in Reduction 10. Moreover, an application of Reduction 10 decreases the number of taxa and the TBR distance both by exactly 1.

6. A win-win scenario

In this section we explore the interplay of sides of a generator that are adjacent to each other. We will see that a generator side whose adjacent sides are densely decorated with taxa triggers reduction rules and that a generator side that does not trigger a reduction rule has adjacent sides that are, on average, only sparsely decorated with taxa. To this end, we establish several results that pinpoint when a subset of taxa is eligible for Operation P or Reduction 10. We begin with a key insight.

Observation 6.1. *Let T and T' be two phylogenetic trees on X that cannot be reduced under Reductions 1–7. Let G be the generator underlying a phylogenetic network N on X such that $r(N) = d_{\text{TBR}}(T, T')$, and let S be a side of G. If at least three taxa are attached to S in obtaining N from G, then T and T' have a CPT-eligible chain unless S is a $1|1|1$ side.*

Proof. If S is a 0-breakpoint side, then it immediately follows that T and T' have a common 3-chain that is CPT eligible. Suppose that S is a 1-breakpoint side. Then S is a $n_1|n_2$ side, where $n_1 \geq 0$ and $n_2 \geq 0$ denote the number of taxa attached to S on either side of the breakpoint. Since $n_1 + n_2 \geq 3$, either $n_1 \geq 2$ or $n_2 \geq 2$. Hence T and T' have a common 2-chain that is pendant in one of T and T' and therefore CPT eligible. Lastly, suppose that S is a 2-breakpoint side. Similar to the 1-breakpoint case, S is a $n_1|n_2|n_3$ side, where $n_1 \geq 0$, $n_2 \geq 0$, and $n_3 \geq 0$ denote the number of taxa attached to S before the first breakpoint, after the first and before the second breakpoint, and after the second breakpoint, respectively. Since $n_1 + n_2 + n_3 \geq 3$ and S is not a $1|1|1$ side it again follows that T and T' have a common 2-chain that is pendant in one of T and T' and therefore CPT eligible. □

In the remainder of this section, we carefully analyze $2|2$ and $2|1|1$ sides, and establish sufficient conditions under which a subset of taxa that decorates such a side is eligible for Operation P or Reduction 10. Let S be a side of a generator G. Viewing G as a graph, S can either be a *simple edge*, i.e. an edge that is not part of a multi-edge, an edge that is part of

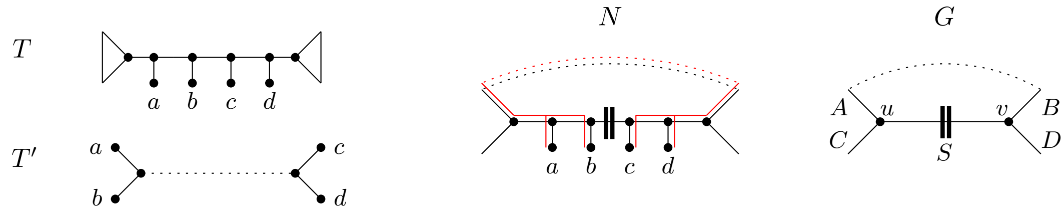

FIGURE 8. The situation described in Lemma 6.2, which concerns 2|2 sides $S = ab|cd$, where S is not part of a multi-edge. The red edges in N indicate an image of T' and dotted edges indicate paths.

a *multi-edge*, or a *loop*. A multi-edge of G contains at most two edges, due to the fact that each vertex of G has degree 3. The only exception is if G has exactly two vertices, and one multi-edge consisting of three edges. This implies that $d_{\text{TBR}} \leq 2$. By assuming throughout the rest of the paper that $d_{\text{TBR}} \geq 3$ we can exclude this case[3]. The following analyses depend on whether S is a simple edge, an edge of a multi-edge, or a loop.

Observe that a 2|2 (or a 1|3) side cannot be a loop because the phylogenetic tree that does not have a breakpoint on that side would contain a cycle.

6.1. 2|2 **sides.**

The proofs of Lemmas 6.2 and 6.3 contain extra anchor points (such as "[Alg 1.CI]") which are used later to help establish the correctness of Algorithm 1.

Lemma 6.2. *Let T and T' be two phylogenetic trees on X that cannot be reduced under Reductions 1–8. Let G be the generator underlying a phylogenetic network N on X that displays T and T' such that $r(N) = d_{\text{TBR}}(T, T')$. Furthermore, let $S = ab|cd$ be a side of G that is a simple edge $\{u, v\}$. Let A, B, C, and D be the four sides incident with S such that A and C are both incident with u, and B and D are both incident with v. If each of A and C is decorated with at least two taxa, or each of B and D is decorated with at least two taxa in obtaining N from G, then $\{a, b, c, d\}$ is eligible for Operation P.*

Proof. Assume without loss of generality that T' has cherries $\{a, b\}$ and $\{c, d\}$ and that T has a non-pendant chain (a, b, c, d) as illustrated in Figure 8. Then (a, b) and (c, d) are two common 2-chains of T and T'. Both of these chains are pendant in T' and therefore CPT eligible. For each $Y \in \{A, B, C, D\}$, let P_Y be the path associated with Y in N. Now, consider an image I of T' in N. Let P be the path from a to c in I. Since S is a 1-breakpoint side, one of P_A and P_C is a subpath of P and, similarly, one of P_B and P_D is a subpath of P. We assume without loss of generality that P_A and P_B are both subpaths of P. It follows that T' has no breakpoint on A or B and, hence each of A and B has at most one breakpoint relative to T. Moreover, by the assumption in the statement of the lemma, at least one of A and B is decorated with at least two taxa in the process of obtaining N from G.

[3]This does not harm the final upper bound $9k-8$ on the size of the kernel, because we can easily test in polynomial time whether $d_{\text{TBR}}(T, T') \leq 2$ and if so exactly compute d_{TBR} in the same time bound. Subsequently we can output a trivial YES/NO instance to complete the kernelization.

Suppose that A is decorated with at least three taxa. Since A has at most one breakpoint, it follows from Observation 6.1 that T and T' have a CPT-eligible chain Z whose leaves are attached to A in obtaining N from G [Alg 1.CI]. Let $K = \{\{a,b\},\{c,d\},Z\}$. By applying the CPT to K, there exists a maximum agreement forest F for T and T' such that each chain in K is preserved in F. Assume that there exists a component U in F such that $\{a,b,c,d\} \subseteq U$. Let U' be the component in F such that $Z \subseteq U'$. If $U \neq U'$, then $T'[U]$ and $T'[U']$ are not vertex-disjoint, a contradiction. Hence $U = U'$. But then $T|(\{a,b,c,d\} \cup Z\}) \neq T'|(\{a,b,c,d\} \cup Z\})$, another contradiction. It follows that U does not exist and $\{a,b,c,d\}$ is eligible for Operation P. An identical analysis holds for when B is decorated with at least three taxa.

We can now assume that neither A nor B is decorated with at least three taxa. Then, by the statement of the lemma, A or B is decorated with exactly two taxa. We establish the lemma for when A is decorated with exactly two taxa. An analogous and symmetric argument holds for when B is decorated with exactly two taxa [Alg 1.CII(c)-(d)]. Let e and f be the two taxa that are attached to A in obtaining N from G such that the path from p_e to p_a in T' does not pass through p_f. Intuitively, e is closer than f to a in N. If $A = |ef$ or $A = ef|$, then T and T' have a common 2-chain (e,f) that is pendant in T. By applying an argument that is similar to that of the last paragraph and setting $Z = \{e,f\}$, we deduce that $\{a,b,c,d\}$ is eligible for Operation P [Alg 1.CI]. Hence $A = ef$ or $A = e|f$. Let F be a maximum agreement forest for T and T'. Assume that there exists a component U in F such that $\{a,b,c,d\} \subseteq U$. Since $T|U = T'|U$, we have $U = \{a,b,c,d\}$. Furthermore, each of e and f is a singleton in F. We now consider two cases, depending on whether A has one or zero breakpoints, and show that there exists another maximum agreement forest for T and T' that has the desired properties such that $\{a,b,c,d\}$ is eligible for Operation P.

First, suppose that $A = ef$ [Alg 1.CII(a)]. Let

$$F' = (F \setminus \{U, \{e\}, \{f\}\}) \cup \{\{a,b,e,f\},\{c,d\}\}$$

be a forest. Noting that $|F'| < |F|$, it follows by the maximality of F that F' is not an agreement forest for T and T'. Hence, by construction of F', there exists a component U' in $F' \setminus \{\{a,b,e,f\}\}$ such that $T[U']$ uses the edge $\{p_e,p_f\}$ in T. Let U'_1, U'_2 be a bipartition of U' such that neither $T[U'_1]$ nor $T[U'_2]$ uses the edge $\{p_e,p_f\}$ in T. As U' is also a component of F, it now follows that

$$F'' = (F \setminus \{U, U', \{e\}, \{f\}\}) \cup \{\{a,b,e,f\},\{c,d\},U'_1,U'_2\}$$

is an agreement forest for T and T' with $|F| = |F''|$ and in which $\{a,b\}$ and $\{c,d\}$ are both preserved, and $\{a,b,c,d\}$ is not preserved. Hence, $\{a,b,c,d\}$ is eligible for Operation P.

Second, suppose that $A = e|f$ [Alg 1.CII(b)]. Observe that (e,a,b,c,d) is a chain of T. Let

$$F' = (F \setminus \{B, \{e\}\}) \cup \{\{a,b,e\},\{c,d\}\}$$

be a forest. Since there exists no component in $F \setminus \{U\}$ whose embedding in T uses p_e, F' is an agreement forest for T and T'. As $|F'| = |F|$ it now follows again that $\{a,b,c,d\}$ is eligible for Operation P. □

Lemma 6.3. *Let T and T' be two phylogenetic trees on X that cannot be reduced under Reductions 1–8. Let G be the generator underlying a phylogenetic network N on X that displays T and T' such that $r(N) = d_{\mathrm{TBR}}(T,T')$. Furthermore, let $S = ab|cd$ be a side of G*

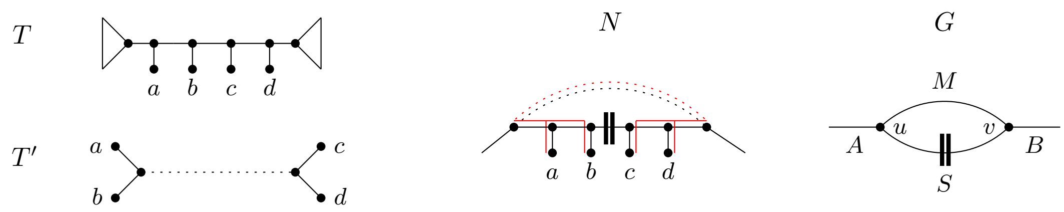

FIGURE 9. The situation described in Lemma 6.3, which concerns 2|2 sides $S = ab|cd$, where S is part of a multi-edge. The red edges in N indicate an image of T' that passes through the path in N that is associated with M and dotted edges indicate paths.

that is part of a multi-edge $\{u, v\}$. Let A, B, and M be the three sides incident with S such that M is incident with u and v, A is only incident with u, and B is only incident with v. Then $\{a, b, c, d\}$ is eligible for Operation P if each of the following conditions hold in obtaining N from G:

(1) one of A and B is decorated with at least two taxa; and
(2) M is decorated with at least one taxon.

Proof. Assume without loss of generality that T' has cherries $\{a, b\}$ and $\{c, d\}$ and that T has a non-pendant chain (a, b, c, d) as illustrated in Figure 9. Then (a, b) and (c, d) are two common 2-chains of T and T'. Both of these chains are pendant in T' and therefore CPT eligible. For each $Y \in \{A, B, M\}$, let P_Y be the path associated with Y in N. Now, consider an image I of T' in N. Let P be the path from a to c in I. Then either P_A and P_B are subpaths of P, or P_M is a subpath of P. If P_A and P_B are subpaths of P then, since the first condition in the statement of the lemma is satisfied, we can apply the same argument as in the proof of Lemma 6.2 to establish that $\{a, b, c, d\}$ is eligible for Operation P.

We may therefore assume that P_M is a subpath of P. Since T does not contain a cycle and the breakpoint on S is relative to T', it follows that M has a single breakpoint that is relative to T.

Let Z be the set of taxa that is attached to M in obtaining N from G. First, assume that Z is CPT eligible [Alg 1.CI]. Then $|Z| \geq 2$, and there exists a maximum agreement forest F for T and T' that preserves each chain in $\{\{a, b\}, \{c, d\}, Z\}$. If there exists a component U in F such that $\{a, b, c, d\} \subseteq U$, then U also contains Z since, otherwise, $T'[U]$ and $T'[U']$ are not vertex-disjoint, where U' is the component in $F \setminus \{U\}$ such that $Z \subseteq U'$. Hence $U = U'$, thereby implying that $T[U] \neq T'[U]$, a contradiction. It now follows that $\{a, b, c, d\}$ is not a subset of any component in F and, thus, $\{a, b, c, d\}$ is eligible for Operation P. Second, assume that Z is not CPT eligible. Since the second condition in the statement of the lemma is satisfied, it follows from the fact that M is a 1-breakpoint side and from the contrapositive of Observation 6.1 that $1 \leq |Z| \leq 2$. Let F be a maximum agreement forest that preserves $\{a, b\}$ and $\{c, d\}$. Again assume that there exists a component U in F such that $\{a, b, c, d\} \subseteq U$. Since $T|U = T'|U$, it follows that $U = \{a, b, c, d\}$. We next consider two cases.

First suppose that $Z = \{e, f\}$. Since Z is not CPT eligible, it follows that $M = e|f$ [Alg 1.CII(e)]. Without loss of generality, we assume that the path from p_a to p_e in T' does not pass through p_f. Observe that (e, a, b, c, d, f) is a chain of T. Moreover, $\{e\}$ and $\{f\}$ are components in F because p_e and p_f are vertices of $T'[U]$. Now, let

$$F' = (F \setminus \{\{U, \{e\}, \{f\}\}) \cup \{\{a, b, e\}, \{c, d, f\}\}.$$

As F is an agreement forest for T and T' and each edge of P is used by $T'[U]$, F' is such a forest as well, contradicting the minimality of F. Hence U does not exist in F and $\{a, b, c, d\}$ is therefore eligible for Operation P.

Second suppose that $Z = \{e\}$. Then $M = |e$ or $M = e|$. Since $T|U = T'|U$, it follows that $\{e\}$ is a component in F. Observe that either (e, a, b, c, d) or (a, b, c, d, e) is a chain of T. Now, if (e, a, b, c, d) is a chain in T [Alg 1.CII(f)], let

$$F' = (F \setminus \{U, \{e\}\}) \cup \{\{a, b, e\}, \{c, d\}\}$$

and, if (a, b, c, d, e) is a chain in T [Alg 1.CII(g)], let

$$F' = (F \setminus \{U, \{e\}\}) \cup \{\{a, b\}, \{c, d, e\}\}.$$

As F is a maximum agreement forest for T and T', it follows that, regardless of which case applies, F' is also such a forest. Thus, $\{a, b, c, d\}$ is eligible for Operation P. □

6.2. 2|1|1 **sides.**

The proofs of Lemmas 6.4, 6.5 and 6.6 contain extra anchor points (such as "[Alg 2.CI]") which are used later to help establish the correctness of Algorithm 2.

Lemma 6.4. *Let T and T' be two phylogenetic trees on X that cannot be reduced under Reductions 1–7. Let G be the generator underlying a phylogenetic network N on X that displays T and T' such that $r(N) = d_{\mathrm{TBR}}(T, T')$. Furthermore, let $S = ab|c|d$ be a side of G that is a simple edge $\{u, v\}$. Let A, B, C, and D be the four sides incident with S such that A and C are both incident with u, and B and D are both incident with v. If each of A and C is decorated with at least two taxa, or each of B and D is decorated with at least two taxa in obtaining N from G, then $\{a, b, c, d\}$ is eligible for Reduction 10.*

Proof. Assume without loss of generality that T has cherries $\{a, b\}$ and $\{c, d\}$ and that T' has a pendant 3-chain (a, b, c) with cherry $\{b, c\}$ as illustrated in Figure 10. Then the 2-chain (a, b) is CPT eligible. Let F be a maximum agreement forest for T and T' such that $\{a, b\}$ is preserved in F. Let U be the component in F with $\{a, b\} \subseteq U$. Then either $\{c\} \in F$ or $\{a, b, c\} \subseteq U$. Assume that the latter holds. Then, as $T|U = T|U'$, we have $U = \{a, b, c\}$. We freely use this observation throughout the rest of the proof.

Now, for each $Y \in \{A, B, C, D\}$, let P_Y be the path associated with Y in N. Furthermore, let I be an image of T in N, and let P be the path from a to c in I. Since S is a 2-breakpoint side, either P_A or P_C is a subpath of P and, similarly, one of P_B or P_D is a subpath of P. We assume without loss of generality that P_A and P_B are both subpaths of P. It follows that T has no breakpoint on A or B and, hence each of A and B has at most one breakpoint relative to T.

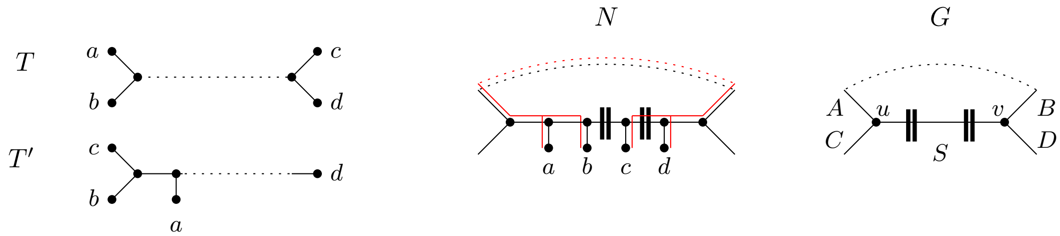

FIGURE 10. The situation described in Lemma 6.4, which concerns $2|1|1$ sides $S = ab|c|d$, where S is not part of a multi-edge. The red edges in N indicate an image of T and dotted edges indicate paths.

Suppose that A is a 1-breakpoint side $A = |ef$ or $A = ef|$ that is decorated with exactly two taxa e and f, or that A is decorated with at least three taxa. Since A has at most one breakpoint, it follows from Observation 6.1 that T and T' have a CPT-eligible chain Z whose leaves are attached to A in obtaining N from G [Alg 2.CI]. Let $K = \{\{a, b\}, Z\}$. By applying the CPT to K, there exists a maximum agreement forest F' for T and T' such that each chain in K is preserved in F'. Let U and U' be the components of F' such that $\{a, b\} \subseteq U$ and $Z \subseteq U'$. Assume that $\{c\} \notin F'$. Then, by the observation in the first paragraph of the proof, we have $U = \{a, b, c\}$. Since $T[U]$ and $T[U']$ are vertex-disjoint, it follows that $U = U'$. In turn, this implies that $T|(\{a, b, c\} \cup Z) \neq T'|(\{a, b, c\} \cup Z)$, thereby contradicting that F' is an agreement forest for T and T'. Hence $\{c\} \in F'$ and, so, $\{a, b, c, d\}$ is eligible for Reduction 10. An identical analysis holds for when B is decorated with at least three taxa [Alg 2.CI]. Hence, one of A and B is decorated with exactly two taxa.

Now, reconsider F. If $\{c\} \in F$, then $\{a, b, c, d\}$ is clearly eligible for Reduction 10. We may therefore assume that $U = \{a, b, c\}$ and, consequently, $\{d\} \in F$. We next distinguish two cases and show that there always exists another maximum agreement forest for T and T' that has the desired property that $\{a, b, c, d\}$ is eligible for Reduction 10.

(1) Suppose that A is decorated with exactly two taxa e and f such that the path from p_e to p_a in T does not pass through p_f. By the definition of an agreement forest, it follows that $\{e\}$ and $\{f\}$ are components of F. Recall that A has at most one breakpoint and that this breakpoint, if it exists, is relative to T'. Hence A is either a 0-breakpoint side $A = ef$ [Alg 2.CII(a)] or a 1-breakpoint side $A = e|f$ [Alg 2.CII(b)]. First, if $A = ef$, let

$$F' = (F \setminus \{U, \{e\}, \{f\}\}) \cup \{\{a, b, e, f\}, \{c\}\}$$

be a forest. Noting that $|F'| < |F|$, it follows by the maximality of F that F' is not an agreement forest for T and T'. Hence, by construction of F', there exists a component U' in $F' \setminus \{\{a, b, e, f\}\}$ such that $T'[U']$ uses the edge $\{p_e, p_f\}$ in T'. Let U'_1, U'_2 be a bipartition of U' such that neither $T'[U'_1]$ nor $T'[U'_2]$ uses the edge $\{p_e, p_f\}$ in T'. As U' is also a component of $F \setminus \{U, \{e\}, \{f\}\}$, it now follows that

$$F'' = (F \setminus \{U, U', \{e\}, \{f\}\}) \cup \{\{a, b, e, f\}, \{c\}, U'_1, U'_2\}$$

is another maximum agreement forest for T and T' in which $\{c\}$ is a singleton. Hence, $\{a,b,c,d\}$ is eligible for Reduction 10. Second, if $A = e|f$, let

$$F' = (F \setminus \{U, \{e\}\}) \cup \{\{a,b,e\}, \{c\}\}$$

be a forest. Since there exists no component in $F \setminus \{U\}$ whose embedding in T' uses p_e, F' is another maximum agreement forest for T and T'. It follows again that $\{a,b,c,d\}$ is eligible for Reduction 10.

(2) Suppose that B is decorated with exactly two taxa e and f such that the path from p_e to p_d in T does not pass through p_f. As in Case (1), $\{e\}$ and $\{f\}$ are components of F. Moreover, B is either a 0-breakpoint side $B = ef$ [Alg 2.CII(c)] or a 1-breakpoint side $B = e|f$ [Alg 2.CII(d)], where the breakpoint is relative to T'. First, if $B = ef$, let

$$F' = (F \setminus \{U, \{d\}, \{e\}, \{f\}\}) \cup \{\{a,b\}, \{c\}, \{d,e,f\}\}$$

be a forest. Noting that $|F'| < |F|$, it follows by the maximality of F that F' is not an agreement forest for T and T'. Hence, by construction of F', there exists a component U' in $F' \setminus \{\{d,e,f\}\}$ such that $T'[U']$ uses the edge $\{p_e, p_f\}$ in T'. Let U'_1, U'_2 be a bipartition of U' such that neither $T'[U'_1]$ nor $T'[U'_2]$ uses the edge $\{p_e, p_f\}$ in T'. As U' is also a component of $F \setminus \{U, \{d\}, \{e\}, \{f\}\}$, it now follows that

$$F'' = (F \setminus \{U, U', \{d\}, \{e\}, \{f\}\}) \cup \{\{a,b\}, \{c\}, \{d,e,f\}, U'_1, U'_2\}$$

is another maximum agreement forest for T and T' in which $\{c\}$ is a singleton. Hence, $\{a,b,c,d\}$ is eligible for Reduction 10. Second, if $B = e|f$, let

$$F' = (F \setminus \{U, \{d\}, \{e\}\}) \cup \{\{a,b\}, \{c\}, \{d,e\}\}.$$

Since there exists no component in $F \setminus \{U\}$ whose embedding in T' uses p_e, F' is another maximum agreement forest for T and T'. Thus $\{a,b,c,d\}$ is eligible for Reduction 10.

□

Lemma 6.5. *Let T and T' be two phylogenetic trees on X that cannot be reduced under Reductions 1–7. Let G be the generator underlying a phylogenetic network N on X that displays T and T' such that $r(N) = d_{\mathrm{TBR}}(T,T')$. Furthermore, let $S = ab|c|d$ be a side of G that is part of a multi-edge $\{u,v\}$. Let A, B, and M be the three sides incident with S such that M is incident with u and v, A is only incident with u, and B is only incident with v. Then $\{a,b,c,d\}$ is eligible for Reduction 10 if each of the following conditions hold in obtaining N from G:*

(1) one of A and B is decorated with at least two taxa; and
(2) M is decorated with at least one taxon.

Proof. Assume without loss of generality that T has cherries $\{a,b\}$ and $\{c,d\}$ and that T' has a pendant 3-chain (a,b,c) with cherry $\{b,c\}$ as illustrated in Figure 11. Then the 2-chain (a,b) is CPT eligible. Let F be a maximum agreement forest for T and T' such that $\{a,b\} \subseteq U$ for some component U in F. As in the proof of Lemma 6.4, we can assume that, if $\{c\} \notin F$, then $U = \{a,b,c\}$.

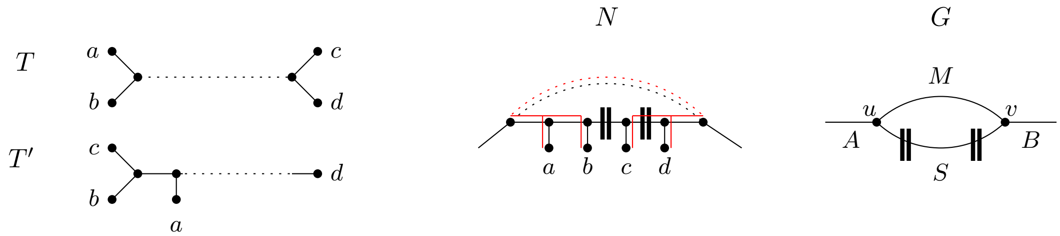

Figure 11. The situation described in Lemma 6.5, which concerns $2|1|1$ sides $S = ab|c|d$, where S is part of a multi-edge. The red edges in N indicate an image of T that uses the path in N that is associated with M and dotted edges indicate paths.

Now, for each $Y \in \{A, B, M\}$, let P_Y be the path associated with Y in N. Furthermore, let I be an image of T in N, and let P be the path from a to c in I. Since S is a 2-breakpoint side, either P_A and P_B is a subpath of P, or P_M is a subpath of P. If P_A and P_B are subpaths of P then, since the first condition in the statement of the lemma is satisfied, we can apply the same argument as in the proof of Lemma 6.4 to establish that $\{a, b, c, d\}$ is eligible for Reduction 10. We may therefore assume that P_M is a subpath of P. As M does not have a breakpoint relative to T, M is a 0-breakpoint side or a 1-breakpoint side in which case the breakpoint is relative to T'.

Suppose that M is a 1-breakpoint side $M = |ef$ or $M = ef|$ that is decorated with exactly two taxa e and f, or that M is decorated with at least three taxa. Since M has at most one breakpoint, it follows from Observation 6.1 that T and T' have a CPT-eligible chain Z whose leaves are attached to M in obtaining N from G [Alg 2.CI]. Applying the same argument as in the third paragraph of the proof of Lemma 6.4 establishes that $\{a, b, c, d\}$ is eligible for Reduction 10.

Since the second condition in the statement of the lemma holds, we complete the proof by considering two cases depending on whether M is decorated with one or two taxa. For both cases, reconsider F and assume that $U = \{a, b, c\}$. We will see that there exists another maximum agreement forest for T and T' that has the desired property that $\{a, b, c, d\}$ is eligible for Reduction 10.

First suppose that M is decorated with only a single taxon e. Clearly, $\{d\}$ and $\{e\}$ are components of F. If M is a 0-breakpoint side [Alg 2.CII(g)], let

$$F' = (F \setminus \{U, \{d\}, \{e\}\}) \cup \{\{a, b, d, e\}, \{c\}\}$$

be a forest. Since $|F'| < |F|$, it follows from the maximality of F that F' is not an agreement forest for T and T'. Hence there exists a component U' in F' (as well as in F) such that $T'[U']$ uses the two edges f and f' that are both incident with p_e and not incident with e. Let U'_1, U'_2 be a bipartition of U' such that neither $T'[U'_1]$ nor $T'[U'_2]$ uses f or f'. Then

$$F'' = (F \setminus \{U, U', \{d\}, \{e\}\}) \cup \{\{a, b, d, e\}, \{c\}, U'_1, U'_2\}$$

is an agreement forest for T and T' with $|F''| = |F|$ and, hence, $\{a, b, c, d\}$ is eligible for Reduction 10. On the other hand, if M is a 1-breakpoint side, then either (c, b, a, e) [Alg

2.CII(h)] or (d, e) [Alg 2.CII(i)] is a pendant chain of T'. In the former case, let

$$F' = (F \setminus \{U, \{e\}\}) \cup \{\{a, b, e\}, \{c\}\}$$

be a forest and, in the latter case let

$$F' = (F \setminus \{U, \{e\}, \{d\}\}) \cup \{\{a, b\}, \{c\}, \{d, e\}\}$$

be a forest. Regardless which applies, F' is an agreement forest and, again, $\{a, b, c, d\}$ is eligible for Reduction 10.

Second suppose that M is decorated with exactly two taxa e and f such that the path from p_a to p_e in T does not pass through p_f. Clearly, $\{e\}$ and $\{f\}$ are components of F. If M is a 0-breakpoint side [Alg 2.CII(e)], let

$$F' = (F \setminus \{U, \{e\}, \{f\}\}) \cup \{\{a, b, e, f\}, \{c\}\}$$

be a forest. As usual, the size of F' contradicts the maximality of F. Hence, there exists a component U' in F' (as well as in F) such that $T'[U']$ uses the edge $\{p_e, p_f\}$. Let U'_1, U'_2 be a bipartition of U' such that neither $T'[U'_1]$ nor $T'[U'_2]$ uses $\{p_e, p_f\}$. Then

$$F'' = (F \setminus \{U, U', \{e\}, \{f\}\}) \cup \{\{a, b, e, f\}, \{c\}, U'_1, U'_2\}$$

is an agreement forest for T and T' with $|F''| = |F|$ and, hence, $\{a, b, c, d\}$ is eligible for Reduction 10.

Finally, if M is a 1-breakpoint side with $M = e|f$ [Alg 2.CII(f)], let

$$F' = (F \setminus \{U, \{e\}\}) \cup \{\{a, b, e\}, \{c\}\}$$

be a forest. As F is an agreement forest for T and T' and (c, b, a, e) is a chain of T', F' is such a forest as well. Thus, $\{a, b, c, d\}$ is eligible for Reduction 10. $\square$

Finally, we turn to loops. Consider two phylogenetic trees T and T' and a phylogenetic network N that displays T and T' such that $r(N) = d_{\mathrm{TBR}}(T, T')$. Let S be a loop side of the generator G that underlies N. Then S is decorated with at least one taxon since, otherwise, there exists a phylogenetic network with strictly fewer than $r(N)$ reticulations that displays T and T'. Moreover if A denotes the side of G that is incident to S then, because T and T' are connected, A has no breakpoint and S has two breakpoints. Hence every loop is adjacent to a 0-breakpoint side. These observations as well as the next lemma, which shows that loop sides exhibit clean behavior in terms of being eligible for Reduction 10, will be convenient for the bounding argument of the next section.

Lemma 6.6. *Let T and T' be two phylogenetic trees on X that cannot be reduced under Reductions 1–7. Let G be the generator underlying a phylogenetic network N on X that displays T and T' such that $r(N) = d_{\mathrm{TBR}}(T, T')$. If $S = ab|c|d$ is a side of G that is a loop, then $\{a, b, c, d\}$ is eligible for Reduction 10.*

Proof. Assume without loss of generality that T has cherries $\{a, b\}$ and $\{c, d\}$ and that T' has a pendant 3-chain (a, b, c) with cherry $\{b, c\}$. Note that, due to the fact that S is a loop, there is at most one subtree between the cherries $\{a, b\}$ and $\{c, d\}$ in T, and at most one subtree between a and d in T' [Alg 2.CII(j)]. Then (a, b) is a CPT-eligible 2-chain. Let F be a maximum agreement forest for T and T' such that $\{a, b\} \subseteq U$ for some component U in F.

Assume that $\{c\} \notin F$. Then, as before, $c \in U$. Since $T|U = T'|U$, we have $U = \{a, b, c\}$ and, therefore, $\{d\} \in F$. It follows that

$$F' = (F \setminus \{U, \{d\}\}) \cup \{\{a, b, d\}, \{c\}\}$$

is an agreement forest for T. Moreover, as $|F'| = |F|$ and $\{c\} \in F'$, $\{a, b, c, d\}$ is eligible for Reduction 10. □

7. Explicit descriptions of Algorithm 1 and 2 for testing eligibility for Operation P or Reduction 10.

Reductions 9 and 10 rely on Algorithms 1 and 2 to test eligibility. These algorithms do not have access to the underlying generator and have to search for the corresponding induced structures in two phylogenetic trees. The algorithms closely mirror the analyses in the proofs of Lemmas 6.2–6.6.

7.1. **Algorithm 1 tests whether $\{a, b, c, d\}$ is eligible for Operation P.**

Algorithm 1 is defined as follows. We first check that T' has cherries $\{a, b\}$ and $\{c, d\}$ and T has a non-pendant chain (a, b, c, d) where a and d are the outermost leaves on the chain. This can easily be verified in polynomial time. If these conditions do not hold, we return NO/DON'T KNOW. Next, if at least one of the following polynomial-time checkable conditions is true, return YES i.e. we conclude that Operation P can be applied. If none of them are true, return NO/DON'T KNOW.

Condition I: In T', the path P from p_a to p_c passes through at least one CPT-eligible chain Z where $Z \cap \{a, b, c, d\} = \emptyset$.
Condition II: Any of situations (a)–(g) from Figure 12 occur.

Crucially, Algorithm 1 does not necessarily detect *all* situations in which $\{a, b, c, d\}$ is eligible for Operation P, and it does not need to either[4]. It only needs to detect *at least* those situations in which Lemmas 6.2 and 6.3 state that Operation P definitely applies. The issue is that these lemmas are stated in terms of an underlying generator G that the algorithm cannot see. The algorithm can only see T and T'. Hence it is necessary to consider, for each subcase considered in the proofs of Lemmas 6.2 and 6.3, the topologies of T and T' that are induced by these subcases. Specifically, to establish the correctness of Algorithm 1, we need show two things: (1) that every subcase considered by the proofs of Lemmas 6.2 and 6.3 induce trees T, T' that cause Algorithm 1 to return YES; (2) that whenever Algorithm 1 returns YES, Operation P applies.

To show (1), we have indicated within the proofs of Lemmas 6.2 and 6.3 which of Conditions I and II(a)-(g) catch each subcase. Condition I captures the parts of those lemmas when the path P, passing through side A, B or M depending on the situation, contains at least one

[4]Indeed, the algorithm returns NO/DON'T KNOW as its failure state because it might still be possible that $\{a, b, c, d\}$ is eligible for Operation P but for reasons beyond those detected by Algorithm 1. However, we do not care about such cases. Functionally speaking a NO/DON'T KNOW answer is therefore interpreted simply as NO. Algorithm 2 has the same behavior.

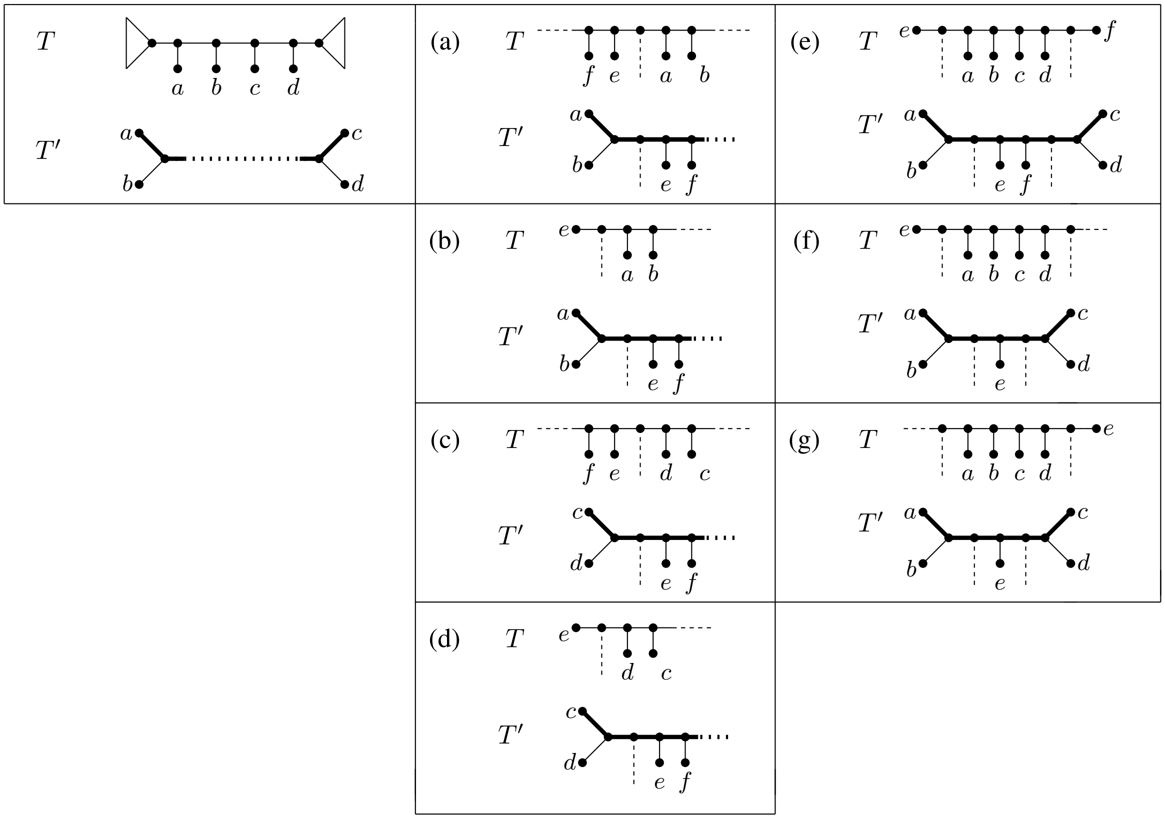

FIGURE 12. Tree topologies checked by Algorithm 1. Path P, as used in Lemmas 6.2 and 6.3, is indicated in bold. Solid lines are edges. Dotted and dashed lines are subtrees that can be optionally present in the tree. Figures (a)–(d) correspond to the situation when the $2|2$ side $ab|cd$ in the underlying generator is a simple edge, and (e)–(g) to when it is a multi-edge.

CPT-eligible chain. Situation (a) of Condition II covers the situation in Lemma 6.2 when the side A has 0 breakpoints and two taxa e and f. Situation (b) is when side A has the form $e|f$. Situations (c) and (d) are symmetrical to (a) and (b): when the path P uses side B rather than A. Situations (e)–(g) concern the cases in Lemma 6.3 where M is a side $e|f$, $e|$ or $|e$ respectively (and the breakpoint is with respect to T). In the depiction of situations (a)-(g) in Figure 12, the dotted and dashed lines correspond to optional subtrees, and these subtrees correspond to sides of the generator that are not directly analyzed in the given subcase. These sides might not be decorated with any taxa, which is why they are optional. For example, in (b) the vertical dashed line corresponds to side C.

To show (2), it is sufficient to focus on those situations when Algorithm 1 returns YES but which do not correspond to a subcase of Lemma 6.2 or 6.3. The only time this can happen, is if Condition I identifies a CPT-eligible chain Z that does not actually lie on side A, B or M. It is still correct in this case to conclude that Operation P is eligible, because exactly the same arguments as used in Lemma 6.2 and 6.3 for CPT-eligible chains Z apply here too.

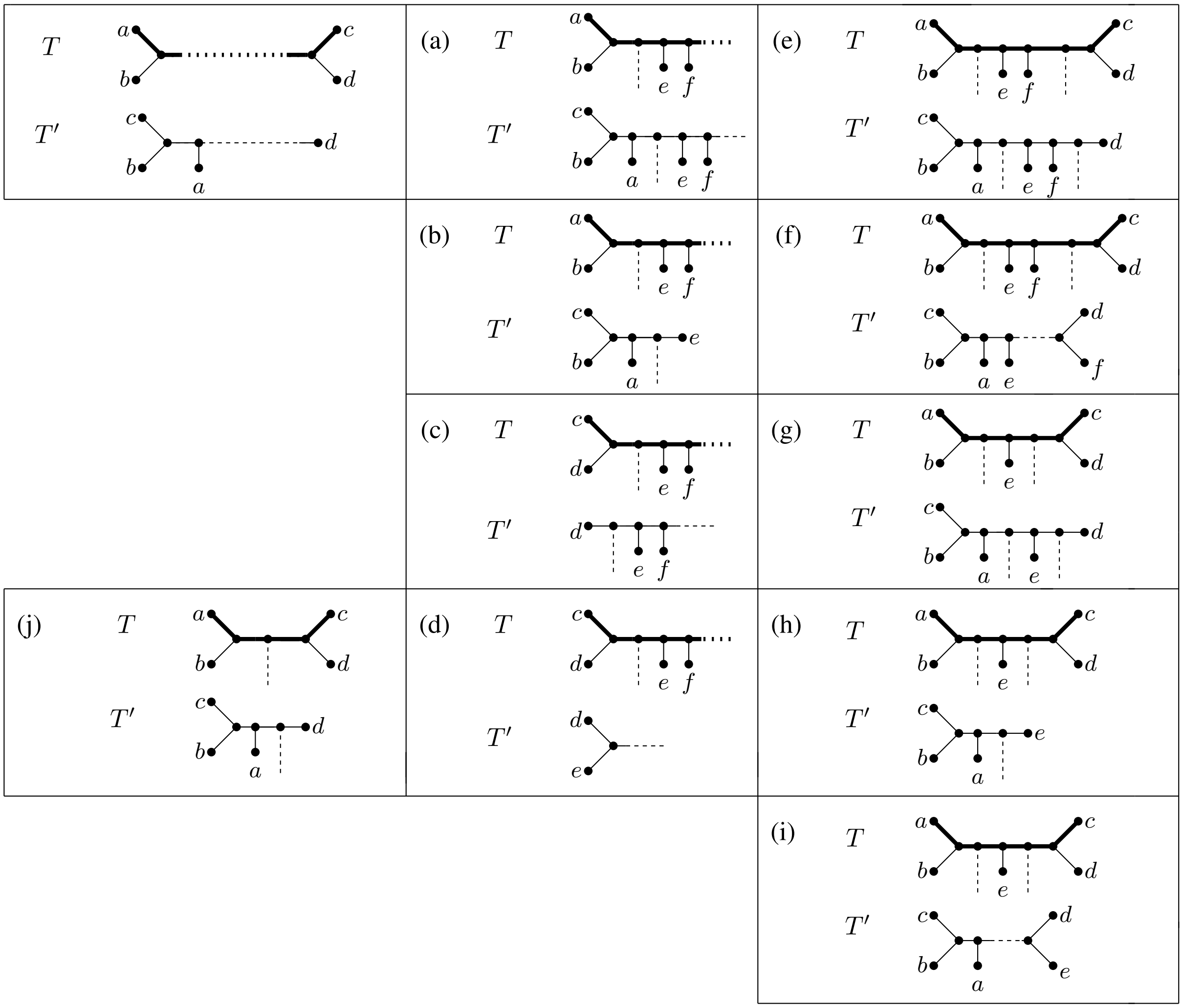

FIGURE 13. Tree topologies checked by Algorithm 2. Path P, as used in Lemmas 6.4 and 6.5, is indicated in bold. Solid lines are edges. Dotted and dashed lines are subtrees that can be optionally present in the tree. Figures (a)–(d) correspond to the situation when the $2|1|1$ side $ab|c|d$ in the underlying generator is a simple edge, and (e)–(i) to when it is a multi-edge. Figure (j) corresponds to Lemma 6.6, which deals with the situation when the side in the underlying generator is a loop.

7.2. **Algorithm 2 tests whether $\{a, b, c, d\}$ is eligible for Reduction 10.**

Algorithm 2 is defined as follows. We first check that T' has a pendant 3-chain (a, b, c) where $\{b, c\}$ is the cherry, and T has two cherries $\{a, b\}$ and $\{c, d\}$. This can easily be confirmed in polynomial time. If these conditions do not hold, we return NO/DON'T KNOW. Next, if at least one of the following polynomial-time checkable conditions is true, return YES i.e. Reduction 10 can be applied. If none of them are true, return NO/DON'T KNOW.

Condition I: In T, the path P from p_a to p_c passes through at least one CPT-eligible chain Z where $Z \cap \{a, b, c, d\} = \emptyset$.
Condition II: Any of situations (a)–(j) from Figure 13 occur.

Analogous to how the correctness of Algorithm 1 is very closely linked to the subcases in Lemmas 6.2 and 6.3, the correctness of Algorithm 2 is very closely linked to the subcases considered in the three Lemmas 6.4, 6.5 and 6.6. Hence, to establish correctness of Algorithm 2 we need to show that all those subcases induce T, T' that cause Algorithm 2 to say YES. Below we describe which subcases each of (a)-(j) are designed to detect, and, in the other direction, within the proofs of the three lemmas we include a reference alongside each subcase to which of (a)-(j) detects it.

Condition I captures the parts of Lemmas 6.4 and 6.5 when the path P, passing through side A, B or M depending on the situation, contains at least one CPT-eligible chain. (As in Algorithm 1, it does not matter if the chain found does not actually lie on A, B or M: it is still correct in this case to conclude that Reduction 10 is eligible.) Situation (a) in Condition II covers the situation in Lemma 6.4 when side A is a 0-breakpoint side with two taxa e and f, and (b) when A is a side $e|f$. Situation (c) covers the situation when side B is a 0-breakpoint side with taxa e and f, and situation (d) when side B is a $e|f$ side; note that situations (c) and (d) are not entirely symmetrical to situations (a) and (b) due to the inherent asymmetry of $2|1|1$ sides. Situations (e)–(i) concern Lemma 6.5. In particular, (e) is when M is a 0-breakpoint side with two taxa e and f and (f) is when M is a 1-breakpoint side $e|f$ (where the breakpoint is with respect to T'). Situation (g) is when M is a 0-breakpoint side with only one taxon e, (h) is when M is a side $e|$, and (i) is when M is a side $|e$. Again, the breakpoints here are with respect to T'. Situation (j) reflects Lemma 6.6.

8. Putting it all together and bounding the size of the kernel

In this section, we establish an improved kernel result for computing the TBR distance that is based on Reductions 1–10. We start by bounding the number of certain types of sides in a generator.

Lemma 8.1. *Let T and T' be two phylogenetic trees on X that cannot be reduced under Reductions 1–10, and let G be a generator that underlies a phylogenetic network N that displays T and T' such that $r(N) = d_{\mathrm{TBR}}(T, T')$. Let s_2 be the number of $2|2$ sides of G, each being decorated with four taxa that are eligible for Operation P, and let s_1 be the number of $1|3$ sides of G. Then $s_1 + s_2 \leq 1$.*

Proof. Suppose that $s_1 + s_2 \geq 2$. By Observation 5.2, we have $s_1 \leq 1$. If $s_1 = 1$ and $s_2 \geq 1$, then T and T' can be reduced by an application of Reduction 9.1. Hence, we may assume that $s_1 = 0$ and $s_2 \geq 2$. Let $S_1 = a'b'|c'd'$ and $S_2 = a''b''|c''d''$ be two $2|2$ sides of G such that each of $\{a', b', c', d'\}$ and $\{a'', b'', c'', d''\}$ are both eligible for Operation P. We establish the lemma by showing that we can apply Reduction 9.2, thereby contradicting that T and T' cannot be reduced under Reductions 1–10. Since $\{a'', b'', c'', d''\}$ is eligible for Operation

P before this operation is applied to $\{a', b', c', d'\}$, note first that S_2 satisfies the conditions in the statement of Lemma 6.2 or 6.3, depending on whether S_2 is part of a multi-edge or not. Now let S and S' be the two phylogenetic trees obtained from T and T', respectively, by applying Operation P to $\{a', b', c', d'\}$. Crucially, by Corollary 5.4, N displays S and S'. As noted in the corollary the only change is that on side S_1 a breakpoint moves slightly. Furthermore, by Lemma 5.3, $d_{\rm TBR}(T, T') = d_{\rm TBR}(S, S')$ which implies that there exists no phylogenetic network that displays S and S' and has strictly fewer than $r(N)$ reticulations. It now follows that $\{a'', b'', c'', d''\}$ is still eligible for Operation P after this operation has been applied to $\{a', b', c', d'\}$ because S_2 still satisfies the conditions in the statement of Lemma 6.2 or 6.3, depending on whether S_2 is part of a multi-edge or not. □

We next use a pessimistic, but safe, counting argument to finally establish Theorem 1.1.

Proof of Theorem 1.1. Let N be a phylogenetic network that displays T and T' such that

$$k = r(N) = d_{\rm TBR}(T, T'),$$

and let G be the generator that underlies N. By [19, Lemma 1], G has $3k - 3$ sides and, by Lemma 4.1, each side of G is decorated with at most four taxa when obtaining N from G. Additionally, by the latter lemma, each side that is decorated with four taxa is a $1|3$, $2|2$, or $2|1|1$ side. Moreover, by Lemma 8.1, the number of $2|2$ sides that are eligible for Operation P plus the number of $1|3$ sides is at most one. We next use the results established in Section 6 to derive the following *adjacency rules* for sides of G that are decorated with four taxa.

- A1. From the contrapositives of Lemmas 6.2 and 6.4, it follows that each side of G (with possibly one exception by Lemma 8.1) that is decorated with four taxa and is not a loop or part of a multi-edge is incident to at least two distinct sides that are each decorated with at most one taxon.
- A2. From the contrapositives of Lemmas 6.3 and 6.5, it follows that each side of G (with possibly one exception by Lemma 8.1) that is decorated with four taxa and is part of a multi-edge is either incident to at least two distinct sides that are not part of the same multi-edge and each decorated with at most one taxon, or the second side in the multi-edge is decorated with zero taxa.
- A3. From the contrapositive of Lemma 6.6, it follows that each side of G that is a loop is decorated with at most three taxa. Furthermore, to avoid disconnecting T or T', each such loop side is incident to a 0-breakpoint side that, by Lemma 4.1(b), is decorated with at most three taxa.

We first deal with the exceptional situation that there is a single side S of G that is decorated with four taxa and does not obey A1 or A2. For the purpose of the upcoming counting argument, we view S as a side that is only decorated with three taxa. This does not affect A1 or A2 because these rules only consider adjacent sides that are decorated with at most one taxon. Furthermore, recalling that S is not a 0-breakpoint side, viewing S as a side that is decorated with three taxa does not affect A3 either. To avoid an underestimate of the final kernel size, we add one to the counting formula below. Next, we consider each side S of G that is decorated with zero or two taxa and view it in one of the following ways for counting purposes. Note that if S is a loop then by the assumed optimality of N it must have at least one taxon.

1. If S is not a loop, not part of a multi-edge, and decorated with zero taxa, we view S as a side that is decorated with one taxon. This does not affect A1–A3.
2. If S is part of a multi-edge and decorated with zero taxa, we view S as a side that is decorated with three taxa and, if subsequently any side S' of G that is incident to S is decorated with four taxa, then we view S' as a side that is decorated with three taxa. This cannot decrease the total number of taxa because S is incident to at most three sides that are each decorated with four taxa. Also, we still obey A1–A3, because any side decorated with four taxa that needed S as a side decorated with zero taxa is now viewed as a side decorated with three taxa.
3. If S is decorated with two taxa, we view S as a side that is decorated with three taxa. Again, this does not affect any of A1–A3.

Now we still have a valid upper bound on the total number of taxa that decorate sides of G, but a simplified counting system because every side is decorated with four, three, or one taxa. Let p and q be the number of sides of G that are decorated with four and three taxa respectively, and let r be the number of sides of G that are decorated with one taxon. Then we have the following optimization problem, where the $+1$ in the objective function is due to the possibly undercounted side decorated with four taxa that is mentioned above and that we view as a side decorated with three taxa.

$$\begin{aligned}\text{maximize } & 4p + 3q + 1r + 1\\ \text{subject to } & p + q + r = 3k - 3\\ & p \leq 2k\\ & r \geq (2/4)p \text{ and}\\ & p, q, r \geq 0 \text{ (and integer)}\end{aligned}$$

The $p \leq 2k$ inequality occurs because, by Lemma 4.1, a side that is decorated with four taxa has at least one breakpoint and there are $2k$ breakpoints in total (i.e., k breakpoints for each tree). Furthermore, each side that is decorated with one taxon can be incident to at most four sides that are each decorated with four taxa. On the other hand, since T and T' cannot be further reduced under any of Reductions 1–10, each side that is decorated with four taxa needs to be incident to at least two sides decorated with one taxon. This implies that $r \geq (2/4)p$. We next substitute $q = (3k-3) - p - r$ and this gives

$$\begin{aligned}\text{maximize } & 9k + p - 2r - 8\\ \text{subject to } & p \leq 2k\\ & r \geq (1/2)p \text{ and}\\ & p, r \geq 0 \text{ (and integer).}\end{aligned}$$

The fact that $r \geq (1/2)p$ implies that the term $(p - 2r)$ in the objective function is at most 0. We conclude that $|X| \leq 9k - 8 = 9d_{\text{TBR}}(T, T') - 8$ is an upper bound on the size of our kernel. $\square$

The bound $9k - 8$ is tight up to an additive term of 1, as the following theorem shows. The additive term is due, in the above analysis, to the at most one generator side with four taxa that does not obey A1 or A2. To establish the next theorem, we need the following definitions. A *binary character* f on X is a function that assigns each element in X to an element in $\{0,1\}$. Let T be a phylogenetic tree on X with vertex set V. An *extension* g of f to V is a function g that assigns each vertex in V to an element in $\{0,1\}$ such that $g(x)=f(x)$ for each $x \in X$. The *parsimony score* of f on T, denoted by $l_f(T)$, denotes the minimum number of edges $\{u,v\}$ in T such that $g(u) \neq g(v)$, ranging over all extensions of f. Now, for two phylogenetic trees T and T' on X, the *maximum parsimony distance on binary characters* d^2_{MP} is defined as $d^2_{\mathrm{MP}}(T,T') = \max_f |l_f(T) - l_f(T')|$ where f ranges over all binary characters on X. It is well-known that $d_{\mathrm{TBR}}(T,T') \geq d^2_{\mathrm{MP}}(T,T')$ [12].

Theorem 8.2. *For each $k \geq 3$ there exist two phylogenetic trees T_k and T'_k with $9k-9$ taxa and $d_{\mathrm{TBR}}(T_k,T'_k) = k$ that cannot be reduced under Reductions 1–10.*

Proof. Let $k \geq 3$. We proceed by building a specific ladder-like generator G_k, converting this to a phylogenetic network N_k, and extracting the trees T_k and T'_k from this. We will then prove that $d_{\mathrm{TBR}}(T_k,T'_k) = k$ and that T_k and T'_k are irreducible under Reductions 1–10.

Generator G_k is built as follows. We take the rectangular $2\times(k+1)$ grid on $2(k+1)$ vertices and suppress the four corner vertices of degree 2. This creates a cubic multigraph G_k with $3(k-1)$ sides. Note that G_k has exactly two pairs of multi-edges. We create N_k by decorating each side of G_k with 3 taxa. Let X_k be the set of all taxa added; we have $|X_k| = 9k-9$. By construction, $r(N_k) = k$. See Figure 14 for the situation $k=5$. Let T_k (respectively, T'_k) be the tree displayed by N_k that is induced by the k solid (respectively, hollow) breakpoints as indicated in the figure. Given that $r(N_k) = k$ we have $d_{\mathrm{TBR}}(T_k,T'_k) \leq k$.

To prove that $d_{\mathrm{TBR}}(T_k,T'_k) \geq k$ we use the same lower-bounding technique as [19, 20]. To this end, it is sufficient to give a binary character f on X_k such that $|l_f(T_k) - l_f(T'_k)| \geq k$. We define f by assigning 0 to each taxon to the left of the red line, as indicated in Figure 14, and assigning 1 to all other taxa, i.e. those to the right of the red line. It is easy to check that $l_f(T_k) = 1$ and that, by Fitch's algorithm or similar [13], $l_f(T'_k) \geq k+1$, so $d_{\mathrm{TBR}}(T_k,T'_k) \geq |l_f(T_k) - l_f(T'_k)| \geq k$ as required. This concludes the proof that $d_{\mathrm{TBR}}(T_k,T'_k) = k$.

Regarding irreducibility, it is helpful to first inventorize some topological features of T_k and T'_k. They have no common pendant subtrees of size 2 or larger, so Reduction 1 is excluded, and they have no common chains of length 4 or longer, so Reduction 2 is excluded. Crucially, each tree has exactly one pendant 3-chain but this is *not* common with the other tree. For T_k this is (p,q,r), where $\{q,r\}$ is its cherry, and for T'_k this is (s,t,u), where $\{t,u\}$ is its cherry. Hence, Reductions 3, 4, 6 and 7 are excluded. Recalling the definition of Reduction 5, we see that if the preconditions for this reduction rule hold, then it also follows that (ℓ_1,ℓ_2,x) is a pendant 3-chain in one tree (with $\{\ell_2,x\}$ the cherry) and (ℓ_1,ℓ_2) is a 2-chain common to both trees. As noted already each tree has exactly one pendant 3-chain. Without loss of generality (due to symmetry between T_k and T'_k), observe that the single pendant 3-chain (p,q,r) in T_k, where $\{q,r\}$ is the cherry, has the property that (p,q) is *not* a 2-chain in T'_k, so Reduction 5 cannot apply.

We now turn to the new reduction rules. Consider Reduction 8, which is only triggered if there is a common 3-chain that is pendant in one tree. Again, this does not exist, so the reduction is excluded. The same fact immediately excludes Reduction 9.1. Reduction 9.2 requires Operation P to execute, and this operation requires one of the trees to have a

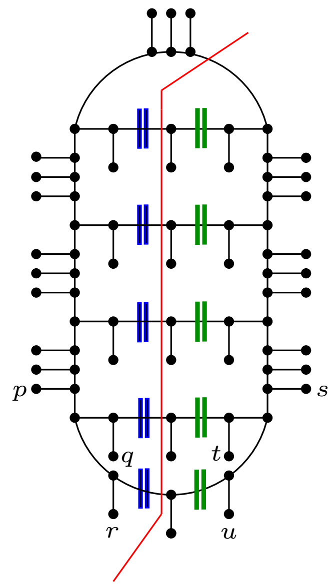

Figure 14. The phylogenetic network N_k for $k = 5$, as constructed in the proof of Theorem 8.2. The k blue double bars represent the breakpoints for the first tree T_k and the k green double bars represent the breakpoints for the second tree T'_k. Taxa p, q, and r form the unique pendant 3-chain of T_k and s, t, and u form the unique pendant 3-chain of T'_k. The red line is used in the proof to show that $d_{\text{TBR}}(T_k, T'_k) \geq k$.

non-pendant 4-chain (a, b, c, d) and the other tree to have cherries $\{a, b\}$ and $\{c, d\}$. Each of T_k and T'_k has exactly $(k + 1)$ cherries, but no pair of these cherries combine to form a 4-chain in the other tree, so Operation P cannot apply. (Viewed from the contrapositive perspective: any 4-chain must contain at least one taxon that is not in a cherry in the other tree.) Finally, consider Reduction 10. The preconditions here require one of T_k and T'_k to have a pendant 3-chain (a, b, c) where $\{b, c\}$ is the cherry, and the other tree to have the cherry $\{a, b\}$. However, the single pendant 3-chain in (without loss of generality) T_k, (p, q, r) where $\{q, r\}$ is the cherry has the property that the first taxon p on the chain is definitely not in a cherry in the other tree, so Reduction 10 cannot execute. We are done. □

9. Conclusion and future work

There are a number of interesting future research directions. The most obvious direction is to design new reduction rules capable of further improving the current $9k-8$ bound. How far below $9k-8$ can we go, and what is the trade off between proof complexity and the obtained decrease in kernel size? Specifically, the existing reduction rules and their associated proofs are already rather complex, requiring extensive auxiliary mathematical machinery and quite some case-checking. It is natural to ask whether the design of further rules and the related proofs can be streamlined and simplified in some fashion by deepening our understanding of the combinatorial behavior of agreement forests. In the meantime (semi-)automated tools for proof verification could be utilized to help keep case-checking under control. We note also that Reduction 8 hints at a wider family of reduction rules. Essentially, it gives us a general recipe for moving certain edges around in the trees such that d_{TBR} is preserved: this allows us to rearrange the trees in such a way that *other* reduction rules are triggered. We expect that such 'indirect' reduction rules will be very useful in the future. Also, the fact that Reduction 8 undoes an application of the chain reduction as an auxiliary step is interesting: it takes a step 'back', in order to move forward. As discussed in [11] this phenomenon merits further study.

Next, an empirical study in the spirit of [30] could investigate how much extra reductive power the new $9k-8$ rules have in practice; the rules for the $11k-9$ kernel do have more practical effect than the $15k-9$ rules. Does this trend continue? Another promising angle to explore is to map the new reduction rules onto other phylogenetic distances to obtain smaller kernels there. This has already been effective in designing strong new reduction rules for the Rooted Subtree Prune and Regraft distance (rSPR) [21]. Strikingly, the complex $11k-9$ reduction rules for TBR distance that formed the inspiration for [21] yielded a much simpler yet comparatively more powerful set of rules for rSPR distance, and also the insight that computation of rSPR distance can actually be modeled as a (cyclic) phylogenetic network construction problem. Indeed, there are many variants of agreement forests in the literature [6], and many applications of generators (see e.g. [28] and references therein), so we are optimistic that ever-more sophisticated reduction rules for TBR will spark advances in kernelization in phylogenetics more widely.

Finally, we note that as the portfolio of reduction rules for a problem expands, complex interactions between the reduction rules can arise. For example, sometimes a newer reduction rule, or combination of reduction rules, subsumes or replaces earlier reduction rules. This phenomenon has been carefully mapped in the context of the intensively-studied Vertex Cover problem (see e.g. the appendix of [11]). As the number of TBR reduction rules increases, it will be insightful to study the way the reduction rules interact.

Acknowledgements. We thank an anonymous referee for their constructive comments and Steve Chaplick for useful discussions. Steven Kelk and Simone Linz were supported by the New Zealand Marsden Fund. Ruben Meuwese was supported by the Dutch Research Council (NWO) KLEIN 1 grant *Deep kernelization for phylogenetic discordance*, project number OCENW.KLEIN.305.

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